MF KVM v2 r7f 驗證報告

來源：v2_r7f/out/release_r7f.log、v2_r7f/fab_r7f/*.txt / *.md、v2_r7f/MUTATION_REPORT.txt（2026-11-09 release，PDF build 20261109-1615）

Gerber `5edbd11614de56c81819c0f49b667f6346e41d8fef006e6f3f2256d9243e99f7`
板檔 `9233a26ad1a1a5a4264edf774e52bf83be37c0ba679b807c393bb46f27976c97`

結論：release 全部閘門通過（`RELEASE ALL GATES PASS`）。這些都是檔案和模型層級的檢查，**沒有任何實體驗證**，見第 2 節。

1. 閘門總表

閘門	原始輸出	結果	重點
ERC（KiCad）	`out/erc.json`、`kicad/` 另跑一次	PASS	KiCad ERC 0；erc_check PASS，2 個已知警告（接受清單內）
DRC（KiCad）	`out/drc.json`	PASS	錯誤 0、未連線 0、parity 0；警告 29（courtyards_overlap 25、starved_thermal 4）
DRC 逐項接受	`drc_instances_r7f.txt`	PASS	29 筆警告全部在逐項接受清單內，清單外 0
verify（電氣、BOM、韌體腳位）	`verify_report.txt`	PASS	RESULT ALL PASS，2 個警告（Y1.4 供電 3.44 V、參考層 0.1 mm 級 9 處）。含 eFuse：ILIM 下限 1.87 A > 1.66 A、dV/dt 斜坡 13.4 ms ≤ 30 ms、EP 錫膏 87.5 %、安裝孔銅環 2.9 mm / 到板邊 0.318 mm
mutation（故意放錯要被抓到）	`MUTATION_REPORT.txt`	PASS	126 次執行，0 次失手
check_case（外殼）	`case_check.txt`	PASS	99/99，含導管 8 項、接地遮蔽與驗收 3 項
hand_solder_gap（U1 旁手焊間隙）	`hand_solder_gap_r7f.txt`	PASS	22 個正面焊盤，最小 0.387 mm（限 0.2）；防焊壩 ≥ 0.1；R80–R81 0.160 mm（限 0.15）
csi_via	`csi_via_r7f.txt`	PASS	非 GND 過孔離 CSI 最近 0.654 mm（限 0.62），沒有例外清單
hotplug（eFuse 輸入）	`hotplug_report.txt`	PASS	VBUS_IN 最高 10.258 V（IN 絕對最大 20 V）、EN 3.280 V（絕對最大 7 V）、U2 VIN ≤ 5.70 V
coldstart（冷啟動）	`coldstart_r7f.txt`	分析	不是閘門。插上瞬間 4.1–5.3 A / 6.5–7.9 µC，不比 USB 參考負載差。全負載在 ≥ 1.5 A 電源起得來；1.0 A、0.55 A 限流埠全負載起不來
impedance	`impedance_r7f.txt`	PASS	IMPEDANCE ALL ON TARGET（場解算，沒有跑 JLC 線上計算器）
s3 解算器自測	`s3_solver_test_r7f.txt`	PASS	RESULT PASS
pairs / skew	`pairs_r7f.txt`	資料	CSI 線內 skew 0.000 / 0.085 / 0.091 mm；TMDS D2 0.003、D1 0.333、D0 0.003、CK 0.096 mm；無 90 度彎
pairgap	`pairgap_r7f.md`	PASS	RESULT PASS
refplane	`refplane_r7f.txt`	PASS	差動線 0.03 mm 內缺口 0；0.1 mm 內 9 處（0.0607 mm²，列出，和 r7e 相同）

maskcheck	`mask_report.txt`	PASS	防焊橋最小 0.1088 mm，< 0.1 mm 0 處；開窗到異網路銅最小 0.1007 mm
holecheck	`hole_report.txt`	PASS	低於 JLC 限值 0 處
bodycheck（零件本體）	`bodycheck_r7f.txt`	PASS	通過；R80 / R81 外形依 EIA 0201 假設，原廠圖沒讀
cpl_pin1	`CPL_polarity_pin1_r7f.md`	PASS	124 列、24 顆要核對。這是檔案層級，U16 等的貼片角度仍要在 JLC 3D 預覽核對
fab_pack	`SHA256.txt`	PASS	Gerber zip sha256 5edbd11614de56c81819c0f49b667f6346e41d8fef006e6f3f2256d9243e99f7

「PASS」是有門檻、會讓 release 失敗的閘門。「分析」「資料」只提供數字，不擋 release。

sha 的已知性質：每次 release 都會重新產生板檔（隨機 UUID、KiCad 存檔順序）且每個 Gerber 有建立時間，所以即使設計沒改，zip 的 sha 也會變。驗收以幾何比對為準：`02b51106` 和 `5edbd116` 的銅、防焊、錫膏 10 層對稱差 0.000000 mm²（附錄 1），team lead 另用自己的解析器比對 15 層相同。`02b51106` 這個檔案作廢。

2. 沒有驗證的項目

- **沒有實體板。** 沒有打過板、沒有焊過、沒有上過電。上面每一項都是檔案或模型層級。
- **沒有 EMI / EMC 測試。**
- **沒有 RF / RSSI 量測。** 天線、U.FL、Wi-Fi 表現都沒有實測。
- **風扇流量沒有驗證。** 導管式的 0.15 L/s 是假設；風扇型號未定，需要有公開 P–Q 曲線、靜壓 ≥ 50 Pa（DELTA 主要-D1：系統阻抗估約 40 Pa）。熱和觸摸溫度全部是模型值。
- **30 ms dV/dt 預算是假設**（取 coldstart 60 ms 模擬窗的一半），不是規格書數字。
- **C70 直流偏壓是假設。** 5 V 下保留 55 %（約 1.21 µF），Murata 的曲線沒有讀到。
- 全負載 1.51 A 是 coldstart 模型算出的斜坡峰值，不是量測。eFuse 在 100 nF 時的開通延遲規格書沒給，沿用 3300 pF 的 289 µs（假設）。
- SI 是線性模型，P4、TC358743、SD 卡、LCD 都沒有原廠 IBIS；不宣稱 HDMI / CSI 協定相容。
- U16 貼片角度、其他 CPL 列的方向要在 JLC 3D 預覽逐顆核對，檔案無法關閉。
- 0201（R80、R81）手焊沒有試過。DELTA 次要-D3：hand_solder_gap 不管 B 面。

3. 附錄清單（原始輸出，原文英文，照原樣排版）

1. 幾何比對：02b51106 對 5edbd116
2. erc_check_report.txt
3. drc_instances_r7f.txt
4. verify_report.txt
5. MUTATION_REPORT.txt
6. case_check.txt
7. hand_solder_gap_r7f.txt
8. csi_via_r7f.txt
9. hotplug_report.txt
10. coldstart_r7f.txt
11. impedance_r7f.txt
12. s3_solver_test_r7f.txt
13. pairs_r7f.txt
14. pairgap_r7f.md
15. refplane_r7f.txt
16. mask_report.txt

17. hole_report.txt
18. bodycheck_r7f.txt（body_report.txt 內容相同）
19. CPL_polarity_pin1_r7f.md
20. CPL_pin1_check.md
21. README.md（打板包說明）
22. SHA256.txt
23. SHA256_extra.txt

附錄 1：幾何比對：02b51106 對 5edbd116

```
p4_kvm_mini-B_Cu.gbr: area old 1389.3920 new 1389.3920 mm2, symmetric difference 0.000000 mm2 in 0 pieces > 1e
-6 mm2; flashes old 678 new 678, only-old 0, only-new 0
p4_kvm_mini-B_Mask.gbr: area old 251.6712 new 251.6712 mm2, symmetric difference 0.000000 mm2 in 0 pieces > 1e
-6 mm2; flashes old 360 new 360, only-old 0, only-new 0
p4_kvm_mini-B_Paste.gbr: area old 186.6763 new 186.6763 mm2, symmetric difference 0.000000 mm2 in 0 pieces > 1
e-6 mm2; flashes old 327 new 327, only-old 0, only-new 0
p4_kvm_mini-F_Cu.gbr: area old 1467.1772 new 1467.1772 mm2, symmetric difference 0.000000 mm2 in 0 pieces > 1e
-6 mm2; flashes old 531 new 531, only-old 0, only-new 0
p4_kvm_mini-F_Mask.gbr: area old 213.2028 new 213.2028 mm2, symmetric difference 0.000000 mm2 in 0 pieces > 1e
-6 mm2; flashes old 213 new 213, only-old 0, only-new 0
p4_kvm_mini-F_Paste.gbr: area old 157.4462 new 157.4462 mm2, symmetric difference 0.000000 mm2 in 0 pieces > 1
e-6 mm2; flashes old 197 new 197, only-old 0, only-new 0
p4_kvm_mini-In1_Cu.gbr: area old 1669.4521 new 1669.4521 mm2, symmetric difference 0.000000 mm2 in 0 pieces >
1e-6 mm2; flashes old 334 new 334, only-old 0, only-new 0
p4_kvm_mini-In2_Cu.gbr: area old 186.5530 new 186.5530 mm2, symmetric difference 0.000000 mm2 in 0 pieces > 1e
-6 mm2; flashes old 334 new 334, only-old 0, only-new 0
p4_kvm_mini-In3_Cu.gbr: area old 1651.9202 new 1651.9202 mm2, symmetric difference 0.000000 mm2 in 0 pieces >
1e-6 mm2; flashes old 334 new 334, only-old 0, only-new 0
p4_kvm_mini-In4_Cu.gbr: area old 1669.4521 new 1669.4521 mm2, symmetric difference 0.000000 mm2 in 0 pieces >
1e-6 mm2; flashes old 334 new 334, only-old 0, only-new 0
```

附錄 2：erc_check_report.txt

```
pin types read back from the schematic: 553 pins {'bidirectional': 73, 'free': 17, 'input': 30, 'open_collecto
r': 1, 'output': 25, 'passive': 344, 'power_in': 57, 'power_out': 5, 'tri_state': 1}
KiCad 10.99.0 ERC: 0 violations; checks ignored by the project: ['four_way_junction', 'simulation_model_issu
e', 'footprint_filter']
open pins: 66
  PWR_FLAG 3V3: L1 to SW / U2.3 power_out
  5V: power_in pins, no source behind a link or connector (needs a power_out pin on the net itself)
  PWR_FLAG AVDD12: FB3 to V1V2 / L2 to SW_1V2 / U15.3 power_out
  PWR_FLAG AVDD25: FB2 to V2V5 / U4.5 power_out
  PWR_FLAG GND: J5.A1B12 external: USB-C GND: the board ground is the 5 V source return
  PWR_FLAG OSC_VDD: FB6 to 3V3 / L1 to SW / U2.3 power_out
  PWR_FLAG VBUS_IN: J5.A4B9 external: USB Type-C vSafe5V 4.75-5.5 V (box power input; Rd only, so no higher vo
ltage is negotiated)
  PWR_FLAG VDDC: FB5 to V1V2 / L2 to SW_1V2 / U15.3 power_out
  PWR_FLAG VDD_MIPI: FB4 to V1V2 / L2 to SW_1V2 / U15.3 power_out
WARN U7.G6 (EDID_SDA, bidirectional) open: EDID_SDA unused. Toshiba TC358743XBG datasheet 2014-05-16 p10 gives
no "when not used" rule for EDID, buffer type N (no pull): floating Schmitt input. Same as the Antmicro refere
nce (pinout audit O2)
WARN U7.H6 (EDID_SCL, bidirectional) open: EDID_SCL unused, as G6
SCHEMATIC sha256 25cea755bd43e5c79691ab5729f4afe6e54523f1d3a74c4c16d147c8d36837a8
ERC_CHECK PASS, 2 warnings
```

附錄 3：drc_instances_r7f.txt

```
warning  courtyards_overlap C10 + U4
warning  courtyards_overlap C19 + C20
warning  courtyards_overlap J4 + U11
warning  courtyards_overlap R42 + R52
warning  courtyards_overlap C20 + U8
warning  courtyards_overlap C10 + Q1
warning  courtyards_overlap Q1 + R43
warning  courtyards_overlap C16 + R26
warning  courtyards_overlap Q1 + R45
warning  courtyards_overlap C13 + C22
warning  courtyards_overlap C13 + C16
warning  courtyards_overlap C41 + R53
warning  courtyards_overlap C17 + U7
warning  courtyards_overlap C16 + R25
warning  courtyards_overlap C16 + R27
warning  courtyards_overlap Q1 + R46
warning  courtyards_overlap L1 + U2
warning  courtyards_overlap C30 + U3
warning  courtyards_overlap Q1 + R44
warning  courtyards_overlap R80 + R81
warning  courtyards_overlap C64 + U2
warning  courtyards_overlap J7 + R78
warning  courtyards_overlap C40 + J7
warning  courtyards_overlap J4 + TP6
warning  courtyards_overlap C21 + U7
warning  starved_thermal U1.52 + zone GND F.Cu
warning  starved_thermal J5.B1A12 + zone GND F.Cu
warning  starved_thermal C14.2 + zone GND B.Cu
warning  starved_thermal R21.1 + zone GND B.Cu
DRC ACCEPT PASS
```

附錄 4：verify_report.txt

```
pads with nets read back 557
PASS every part on the board has its pins in data/pin_types.json
PASS design.json and the board agree pin for pin (557 pads)
PASS r7f: every mounting hole is a plated pad on GND (lid and plate ground through the posts) [('H1.1', 'GN
D'), ('H2.1', 'GND'), ('H3.1', 'GND'), ('H4.1', 'GND')]
PASS r7f: mounting hole copper ring <= 2.9 mm or copper to the board edge >= 0.3 mm (REVIEW_r7f fix kept) [('H
1.1', 2.9, 0.318), ('H2.1', 2.9, 0.318), ('H3.1', 2.9, 0.318), ('H4.1', 2.9, 0.318)]
PASS every fitted resistor has a value and a tolerance decoded from its part number
RAILS (derived, V)
  VBUS_IN    4.7500 - 5.5000   J5.A4B9: USB Type-C vSafe5V 4.75-5.5 V (box power input; Rd only, so no higher
voltage is negotiated)
  HOST_VBUS  4.7500 - 5.5000   J3.A4B9: USB Type-C vSafe5V 4.75-5.5 V (host port VBUS, used for sensing only)
  5V         4.7500 - 5.5000   U16 eFuse output from VBUS_IN: TI TPS2595xx SLVSE57C (Apr 2018), datasheets_r7
f/tps2595.pdf p7 VCLAMP 5.2-5.7 V (TPS25953x), output follows IN below VOVC 5.5-5.9 V
  5V_DAMP    4.7500 - 5.5000   5V through R68 (drop not included here; see the drop terms in the supply rules)
  HDMI_5V    4.7000 - 5.3000   J2.18: HDMI 1.4 +5V Power from the source, 4.7-5.3 V (as used in R7B_REPORT M5;
spec not re-read for r7c)
  3V3        3.1846 - 3.4405   U2 via L1: FB network R50, R51, VREF [0.588, 0.612], IFB +-50 nA
  OSC_VDD    3.1846 - 3.4405   3V3 through FB6 (drop not included here; see the drop terms in the supply rule
s)
  SD_SRC     3.1846 - 3.4405   3V3 through R42 (drop not included here; see the drop terms in the supply rule
s)
  SD1_VDD    3.1846 - 3.4405   SD_SRC through Q1 (drop not included here; see the drop terms in the supply rul
es)
  V2V5       2.4650 - 2.5350   U4 fixed output: TI TLV733P SBVS235C p6 DC output accuracy +-1.4% over temperat
ure
  AVDD25     2.4650 - 2.5350   V2V5 through FB2 (drop not included here; see the drop terms in the supply rule
s)
  V1V2       1.1831 - 1.2373   U15 via L2: FB network R61, R63, R62, R65, R66, VREF [0.588, 0.612], IFB +-50 n
A
  AVDD12     1.1831 - 1.2373   V1V2 through FB3 (drop not included here; see the drop terms in the supply rule
s)
  VDD_MIPI   1.1831 - 1.2373   V1V2 through FB4 (drop not included here; see the drop terms in the supply rule
s)
  VDDC       1.1831 - 1.2373   V1V2 through FB5 (drop not included here; see the drop terms in the supply rule
s)
  V1V2_SNS   1.1831 - 1.2373   V1V2 through R65 (drop not included here; see the drop terms in the supply rule
s)
  GND        0.0000 - 0.0000   J5.A1B12: USB-C GND: the board ground is the 5 V source return
ELECTRICAL RULES (pin, net found, rule, source)
  ok   Q1.1    SD_PWR_N   |VGS| at most 3.44 V against 12 V; gate level with the GPIO released: 0.0 V (must be
defined)  [AOS AO3401A Rev 3.1 p1 VGS +-12 V; a floating gate leaves the SD supply undefined while GPIO45 is h
igh-Z after reset]
  ok   U1.85   3V3        supply 3.0-3.6 V: 3V3 3.1846-3.4405 V  [Waveshare ESP32-P4-Module wiki: 3.3 V suppl
y; 3.0-3.6 is the P4 VDDPST range, ESP32-P4 datasheet v0.7 p32 (not re-read for r7c)]
  ok   U1.86   3V3        supply 3.0-3.6 V: 3V3 3.1846-3.4405 V  [as 85]
  ok   U1.84   3V3        supply 3.0-3.6 V: 3V3 3.1846-3.4405 V  [as 85]
  ok   U1.87   ESP_EN     pulled high: ESP_EN static (3.184, 3.44) V, VIH_nRST 0.75 x VDD = 2.58 V  [Espressif
ESP32-P4 datasheet v0.7 p34 CHIP_PU: do not leave floating; p90 VIH_nRST 0.75 VDD]
  ok   U1.18   HOST_VBUS_DIV static level of HOST_VBUS_DIV 2.824-3.329 V against VIH 2.580 V  [Espressif ESP32
-P4 datasheet v0.7 p89 VIH 0.75 VDD, max VDD+0.3]
  ok   U1.18   HOST_VBUS_DIV highest voltage on HOST_VBUS_DIV 3.329 V (static 3.329, drivers none) against 3.4
85 V (3V3 at its minimum + 0.3)  [Espressif ESP32-P4 datasheet v0.7 p89 VIH 0.75 VDD, max VDD+0.3]
  ok   U14.1   USB2_DP    highest voltage on USB2_DP 3.440 V (static None, drivers ['U1.51 on 3V3']) against
5.5 V (absolute)  [TI TPDxE05U06 SLVSBO7O (Aug 2024), datasheets_r7/tpd4e05u06_slvsbo7o.pdf p7 VRWM 5.5 V]
  ok   U14.2   USB2_DN    highest voltage on USB2_DN 3.440 V (static None, drivers ['U1.50 on 3V3']) against
```

```
5.5 V (absolute)  [TI TPDxE05U06 SLVSBO7O (Aug 2024), datasheets_r7/tpd4e05u06_slvsbo7o.pdf p7 VRWM 5.5 V]
  ok   U15.4   3V3        supply 2.7-5.5 V: 3V3 3.1846-3.4405 V  [Silergy AN_SY8089/A Rev 0.9A p4 VIN 2.7-5.5,
p2 abs 6.0]
  ok   U15.4   3V3        absolute max 6.0 V: 3.440 V  [Silergy AN_SY8089/A Rev 0.9A p4 VIN 2.7-5.5, p2 abs 6.
0]
  ok   U15.1   3V3        static level of 3V3 3.185-3.440 V against VIH 1.500 V  [Silergy AN_SY8089/A Rev 0.9A
p4 VENH 1.5, p2 EN abs VIN+0.6, "do not float"]
  ok   U15.1   3V3        highest voltage on 3V3 3.440 V (static 3.44, drivers none) against 4.04 V (3V3 + 0.6
(same rail, same instant))  [Silergy AN_SY8089/A Rev 0.9A p4 VENH 1.5, p2 EN abs VIN+0.6, "do not float"]
  ok   U16.3   VBUS_IN    supply 2.7-18.0 V: VBUS_IN 4.7500-5.5000 V  [TI TPS2595xx SLVSE57C (Apr 2018), datas
heets_r7f/tps2595.pdf p5 7.3 VIN 2.7-18 V, 7.1 abs 20 V]
  ok   U16.3   VBUS_IN    absolute max 20.0 V: 5.500 V  [TI TPS2595xx SLVSE57C (Apr 2018), datasheets_r7f/tps2
595.pdf p5 7.3 VIN 2.7-18 V, 7.1 abs 20 V]
  ok   U16.2   EF_EN      highest voltage on EF_EN 1.783 V (static 1.783, drivers none) against 6.0 V (absolut
e)  [TI TPS2595xx SLVSE57C (Apr 2018), datasheets_r7f/tps2595.pdf p5 7.3 VEN 0-6 V (7.1 abs 7 V)]
  ok   U2.4    5V         supply 2.7-5.5 V: 5V 4.7500-5.5000 V  [Silergy AN_SY8089/A Rev 0.9A p4 VIN 2.7-5.5,
p2 abs 6.0]
  ok   U2.4    5V         absolute max 6.0 V: 5.500 V  [Silergy AN_SY8089/A Rev 0.9A p4 VIN 2.7-5.5, p2 abs 6.
0]
  ok   U2.1    5V         static level of 5V 4.750-5.500 V against VIH 1.500 V  [Silergy AN_SY8089/A Rev 0.9A
p4 VENH 1.5, p2 EN abs VIN+0.6, "do not float"]
  ok   U2.1    5V         highest voltage on 5V 5.500 V (static 5.5, drivers none) against 6.1 V (5V + 0.6 (sa
me rail, same instant))  [Silergy AN_SY8089/A Rev 0.9A p4 VENH 1.5, p2 EN abs VIN+0.6, "do not float"]
  ok   U3.1    USB_DN     highest voltage on USB_DN 3.440 V (static None, drivers ['U1.48 on 3V3']) against 5.
5 V (absolute)  [TI TPDxE05U06 SLVSBO7O (Aug 2024), datasheets_r7/tpd4e05u06_slvsbo7o.pdf p7 VRWM 5.5 V]
  ok   U3.2    USB_DP     highest voltage on USB_DP 3.440 V (static 0.05, drivers ['U1.49 on 3V3']) against 5.
5 V (absolute)  [TI TPDxE05U06 SLVSBO7O (Aug 2024), datasheets_r7/tpd4e05u06_slvsbo7o.pdf p7 VRWM 5.5 V]
  ok   U4.1    3V3        supply 1.4-5.5 V: 3V3 3.1846-3.4405 V  [TI TLV733P SBVS235C p5 6.3]
  ok   U4.3    3V3        static level of 3V3 3.185-3.440 V against VIH 0.900 V  [TI TLV733P SBVS235C p6 VEN(H
I) 0.9, p5 VEN abs VIN+0.3]
  ok   U4.3    3V3        highest voltage on 3V3 3.440 V (static 3.44, drivers none) against 3.74 V (3V3 + 0.3
(same rail, same instant))  [TI TLV733P SBVS235C p6 VEN(HI) 0.9, p5 VEN abs VIN+0.3]
  ok   U6.1    3V3        supply 0.9-5.5 V: 3V3 3.1846-3.4405 V  [TI TCA9517A SCPS245E (rev. Oct 2025) p5 5.3]
  ok   U6.8    5V         supply 2.7-5.5 V: 5V 4.7500-5.5000 V  [TI TCA9517A SCPS245E (rev. Oct 2025) p5 5.3]
  ok   U6.5    5V         static level of 5V 4.750-5.500 V against VIH 3.850 V  [TI TCA9517A SCPS245E (rev. Oc
t 2025) p5 5.3 VIH EN 0.7 x VCCB, max 5.5]
  ok   U6.5    5V         highest voltage on 5V 5.500 V (static 5.5, drivers none) against 5.5 V (absolute)
[TI TCA9517A SCPS245E (rev. Oct 2025) p5 5.3 VIH EN 0.7 x VCCB, max 5.5]
  ok   U6.2    DDC3_SCL   pull-up from DDC3_SCL to 3V3 (pin 1): ['R25']  [TI TCA9517A SCPS245E (rev. Oct 2025)
p4: A side "connect to VCCA through a pull-up resistor"]
  ok   U6.3    DDC3_SDA   pull-up from DDC3_SDA to 3V3 (pin 1): ['R26']  [TI TCA9517A SCPS245E (rev. Oct 2025)
p4]
  ok   U6.6    DDC_SDA    pull-up from DDC_SDA to 5V (pin 8): ['R28']  [TI TCA9517A SCPS245E (rev. Oct 2025) p
4: B side "connect to VCCB through a pull-up resistor"]
  ok   U6.7    DDC_SCL    pull-up from DDC_SCL to 5V (pin 8): ['R27']  [TI TCA9517A SCPS245E (rev. Oct 2025) p
4]
  ok   U7.B1   3V3        supply 3.135-3.465 V: 3V3 3.1846-3.4405 V  [Toshiba TC358743XBG datasheet 2014-05-16
p15 5.2 AVDD33]
  ok   U7.F3   3V3        supply 3.135-3.465 V: 3V3 3.1846-3.4405 V  [Toshiba TC358743XBG datasheet 2014-05-16
p15 5.2 AVDD33]
  ok   U7.B2   AVDD12     supply 1.15-1.25 V: AVDD12 1.1616-1.2373 V (after AVDD12 drop 21.5 mV)  [Toshiba TC3
58743XBG datasheet 2014-05-16 p15 5.2 AVDD12]
  ok   U7.D3   AVDD12     supply 1.15-1.25 V: AVDD12 1.1616-1.2373 V (after AVDD12 drop 21.5 mV)  [Toshiba TC3
58743XBG datasheet 2014-05-16 p15 5.2 AVDD12]
  ok   U7.B5   AVDD25     supply 2.25-2.75 V: AVDD25 2.4650-2.5350 V  [Toshiba TC358743XBG datasheet 2014-05-1
6 p15 5.2 AVDD25]
  ok   U7.C3   VDDC       supply 1.1-1.3 V: VDDC 1.1331-1.2373 V (after BEAD drop 50.0 mV)  [Toshiba TC358743X
BG datasheet 2014-05-16 p15 5.2 VDDC]
  ok   U7.F5   VDDC       supply 1.1-1.3 V: VDDC 1.1331-1.2373 V (after BEAD drop 50.0 mV)  [Toshiba TC358743X
BG datasheet 2014-05-16 p15 5.2 VDDC]
```

```
  ok   U7.G2   VDDC       supply 1.1-1.3 V: VDDC 1.1331-1.2373 V (after BEAD drop 50.0 mV)  [Toshiba TC358743X
BG datasheet 2014-05-16 p15 5.2 VDDC]
  ok   U7.C6   VDD_MIPI   supply 1.1-1.3 V: VDD_MIPI 1.1331-1.2373 V (after BEAD drop 50.0 mV)  [Toshiba TC358
743XBG datasheet 2014-05-16 p15 5.2 VDD_MIPI]
  ok   U7.F6   VDD_MIPI   supply 1.1-1.3 V: VDD_MIPI 1.1331-1.2373 V (after BEAD drop 50.0 mV)  [Toshiba TC358
743XBG datasheet 2014-05-16 p15 5.2 VDD_MIPI]
  ok   U7.F4   3V3        supply 3.0-3.6 V: 3V3 3.1846-3.4405 V  [Toshiba TC358743XBG datasheet 2014-05-16 p15
5.2 VDDIO1]
  ok   U7.H7   3V3        supply 1.65-3.6 V: 3V3 3.1846-3.4405 V  [Toshiba TC358743XBG datasheet 2014-05-16 p1
5 5.2 VDDIO2]
  ok   U7.E5   GND        must be low: GND 0.000 V against VIL 0.955 V  [Toshiba TC358743XBG datasheet 2014-05
-16 p10 TEST "0: Normal mode", p16 VIL 0.3 VDDIO]
  ok   U7.A3   GND        must be low: GND 0.000 V against VIL 0.0 V  [Toshiba TC358743XBG datasheet 2014-05-1
6 p11 VPGM "please tie to ground"]
  ok   U7.B4   GND        must be low: GND 0.000 V against VIL 0.955 V  [Toshiba TC358743XBG datasheet 2014-05
-16 p10 IR, Sch input with no pull and no "when not used" rule: r7c ties it low (VIL 0.3 VDDIO, p16) so it can
not float]
  ok   U7.H5   TC_REFCLK  driven by Y1.3: VOH 0.9 x OSC_VDD >= 2.866 V against VIH 2.408 V  [Toshiba TC358743X
BG datasheet 2014-05-16 p10 REFCLK on VDDIO2, p16 VIH 0.7 VDDIO, p15 VIN_IO max VDDIO+0.3; YXC YSO110TR (LCSC
C725995) datasheet, datasheets_r7/dl/C725995_ds.pdf p1 VOH 90% VDD, load 15 pF max]
  ok   U7.H5   TC_REFCLK  highest voltage on TC_REFCLK 3.440 V (static None, drivers ['Y1.3 on OSC_VDD']) agai
nst 3.485 V (3V3 at its minimum + 0.3)  [Toshiba TC358743XBG datasheet 2014-05-16 p10 REFCLK on VDDIO2, p16 VI
H 0.7 VDDIO, p15 VIN_IO max VDDIO+0.3]
  ok   U7.G5   TC_RST     highest voltage on TC_RST 3.440 V (static 0.102, drivers ['U1.32 on 3V3']) against
3.485 V (3V3 at its minimum + 0.3)  [Toshiba TC358743XBG datasheet 2014-05-16 p10 RESETN on VDDIO2, p15 VIN_I
O]
  ok   U7.G4   I2C_SDA    highest voltage on I2C_SDA 3.440 V (static 3.44, drivers ['U1.22 on 3V3']) against
3.74 V (3V3 + 0.3 (same rail, same instant))  [Toshiba TC358743XBG datasheet 2014-05-16 p10 I2C on VDDIO2, p15
VIN_IO]
  ok   U7.H4   I2C_SCL    highest voltage on I2C_SCL 3.440 V (static 3.44, drivers ['U1.23 on 3V3']) against
3.74 V (3V3 + 0.3 (same rail, same instant))  [Toshiba TC358743XBG datasheet 2014-05-16 p10 I2C on VDDIO2, p15
VIN_IO]
  ok   U7.G3   DDC3_SDA   highest voltage on DDC3_SDA 3.440 V (static 3.44, drivers ['U6.3 on 3V3']) against
5.5 V (absolute)  [Toshiba TC358743XBG datasheet 2014-05-16 p10 DDC on VDDIO1, p11 Note1 "5V tolerant" (5.5 V
ASSUMED as the meaning of 5 V tolerant)]
  ok   U7.H3   DDC3_SCL   highest voltage on DDC3_SCL 3.440 V (static 3.44, drivers ['U6.2 on 3V3']) against
5.5 V (absolute)  [Toshiba TC358743XBG datasheet 2014-05-16 p10, p11 Note1]
  ok   U7.H2   HPDI       static level of HPDI 4.185-5.300 V against VIH 2.408 V  [Toshiba TC358743XBG datashe
et 2014-05-16 p10 HPDI on VDDIO1, p11 Note1 5 V tolerant, p16 VIH 0.7 VDDIO, IILH 10 uA]
  ok   U7.H2   HPDI       highest voltage on HPDI 5.300 V (static 5.3, drivers none) against 5.5 V (absolute)
[Toshiba TC358743XBG datasheet 2014-05-16 p10 HPDI on VDDIO1, p11 Note1 5 V tolerant, p16 VIH 0.7 VDDIO, IILH
10 uA]
  ok   U7.A1   TC_REXT    2000 ohm +-1% to 3V3: [('R20', 1980, 2020)]  [Toshiba TC358743XBG datasheet 2014-05-
16 p11 REXT "connect to AVDD33 with a 2K ohm resistor (+-1%)"]
  ok   U8.1    D1_N       highest voltage on D1_N 3.440 V (static None, drivers ['U7.E2 on 3V3']) against 5.5
V (absolute)  [TI TPDxE05U06 SLVSBO7O (Aug 2024), datasheets_r7/tpd4e05u06_slvsbo7o.pdf p7 VRWM 5.5 V]
  ok   U8.2    D1_P       highest voltage on D1_P 3.440 V (static None, drivers ['U7.E1 on 3V3']) against 5.5
V (absolute)  [TI TPDxE05U06 SLVSBO7O (Aug 2024), datasheets_r7/tpd4e05u06_slvsbo7o.pdf p7 VRWM 5.5 V]
  ok   U8.4    D2_N       highest voltage on D2_N 3.440 V (static None, drivers ['U7.F2 on 3V3']) against 5.5
V (absolute)  [TI TPDxE05U06 SLVSBO7O (Aug 2024), datasheets_r7/tpd4e05u06_slvsbo7o.pdf p7 VRWM 5.5 V]
  ok   U8.5    D2_P       highest voltage on D2_P 3.440 V (static None, drivers ['U7.F1 on 3V3']) against 5.5
V (absolute)  [TI TPDxE05U06 SLVSBO7O (Aug 2024), datasheets_r7/tpd4e05u06_slvsbo7o.pdf p7 VRWM 5.5 V]
  ok   U9.1    CK_N       highest voltage on CK_N 3.440 V (static None, drivers ['U7.C2 on 3V3']) against 5.5
V (absolute)  [TI TPDxE05U06 SLVSBO7O (Aug 2024), datasheets_r7/tpd4e05u06_slvsbo7o.pdf p7 VRWM 5.5 V]
  ok   U9.2    CK_P       highest voltage on CK_P 3.440 V (static None, drivers ['U7.C1 on 3V3']) against 5.5
V (absolute)  [TI TPDxE05U06 SLVSBO7O (Aug 2024), datasheets_r7/tpd4e05u06_slvsbo7o.pdf p7 VRWM 5.5 V]
  ok   U9.4    D0_N       highest voltage on D0_N 3.440 V (static None, drivers ['U7.D2 on 3V3']) against 5.5
V (absolute)  [TI TPDxE05U06 SLVSBO7O (Aug 2024), datasheets_r7/tpd4e05u06_slvsbo7o.pdf p7 VRWM 5.5 V]
  ok   U9.5    D0_P       highest voltage on D0_P 3.440 V (static None, drivers ['U7.D1 on 3V3']) against 5.5
V (absolute)  [TI TPDxE05U06 SLVSBO7O (Aug 2024), datasheets_r7/tpd4e05u06_slvsbo7o.pdf p7 VRWM 5.5 V]
```

```
  WARN  Y1.4    OSC_VDD    supply 1.8-3.3 V: OSC_VDD 3.1846-3.4405 V  [YXC YSO110TR (LCSC C725995) datasheet,
datasheets_r7/dl/C725995_ds.pdf p1 supply 1.8-3.3 V (listed values, no tolerance given)]
  ok   Y1.1    OSC_VDD    static level of OSC_VDD 3.185-3.440 V against VIH 2.408 V  [YXC YSO110TR (LCSC C7259
95) datasheet, datasheets_r7/dl/C725995_ds.pdf p1 tri-state high 70% VDD]
  ok   Y1.3    OSC_OUT    output load 6.6 pF (fitted none + 6.6 trace/pad/input estimate) against 15 pF  [YXC
YSO110TR (LCSC C725995) datasheet, datasheets_r7/dl/C725995_ds.pdf p1 VOH 90% VDD, load 15 pF max]
PASS electrical rule table: 69 rules from data/pin_types.json, 0 failed
WARN Y1.4: supply 1.8-3.3 V: OSC_VDD 3.1846-3.4405 V [YXC YSO110TR (LCSC C725995) datasheet, datasheets_r7/dl/
C725995_ds.pdf p1 supply 1.8-3.3 V (listed values, no tolerance given)]
PASS every fitted IC, transistor and oscillator is a known part with its own limit table
PASS no net has two push-pull or power outputs (pin types from data/pin_types.json)
PASS every ground pin named in the vendor tables is on board ground (net of J5 GND)
PASS firmware sdkconfig.v2r7_rev3: I2C 7/8, bridge reset 23, LCD 9-13, SD 39-45, ATX 2/4/5 each reach the vend
or pin they drive
PASS r7f: J5 VBUS reaches 5V only through the eFuse ['U16'] (IN VBUS_IN, OUT 5V)
PASS r7f: nothing bypasses the eFuse U16 between VBUS_IN and 5V (any part, either pin order)
PASS r7f: U16 ILM (EF_ILM) has one resistor to GND within 487-5000 ohm (SLVSE57C 7.3), current limit 2.02-2.06
A (Eq. 4) [('R73', '1')] [('R73', 'EF_ILM', 'GND', 990.0, 1010.0)]
PASS r7f: U16 ILIM at its low end 1.87 A carries the full load 1.51 A x 1.1 = 1.66 A (Eq. 4 at R max, ILIM -7.
5 %)
PASS r7f: U16 dVdt (EF_DVDT) has one capacitor to GND of at least 3300 pF (SLVSE57C 7.3) [('C71', '2')]
PASS r7f: U16 dVdt ramp to 5 V 13.4 ms at CdVdt +10 % within the 30 ms start-up budget (Eq. 3; CdVdt 100.0 nF)
PASS r7f: U16 EP paste: two B.Paste windows on the EP, 87.5 % of it (80-95 %, TI DSG0008A stencil), EP pad its
elf without paste (2 windows, inside True)
PASS r7f: U16 EN/UVLO divider from VBUS_IN and to GND, UVLO rising (3.49, 4.03) V within 3.0-4.6 V (VUVLO(R)
1.13-1.27 V, SLVSE57C 7.5) [('R72', '2'), ('R71', '2')]
PASS r7f: HDMI CEC (HDMI_CEC) has a TPD1E10B06 clamp to GND ['D1']
PASS TC358743 INT is open, as the firmware expects (CONFIG_KVM_BRIDGE_INT_GPIO=-1)
PASS ATX PWR and RST opto transistors appear across J4 1-2 and 3-4 (collector, emitter)
PASS CSI-2: 2 lanes and clock, module pin to TC358743 ball by name, P to P and N to N
PASS HDMI connector to TC358743 by function: TMDS, CEC, DDC via TCA9517A B to A side, HPD from HPDO through a
resistor
PASS every TMDS and USB data line has a TPD4E05U06 channel on it
PASS USB-C: A/B sides joined, Rd 5.1k +-20% on each CC, host port on USB_DP/DM, power port on GPIO25/24 (Waves
hare alt names)
PASS host VBUS and box 5 V are separate nets with no link: neither can feed the other (box: ['5V', 'VBUS_IN'])
PASS antenna: module LNA_OUT (pin 2) to the U.FL signal contact
PASS microSD: VDD from Q1 drain, CMD and DAT0-3 pulled up to ESP_LDO_VO4
PASS SD switch source fed from the module 3.3 V rail, not from ESP_LDO_VO4 (reached through links: ['3V3', 'OS
C_VDD', 'SD_SRC', 'SW'])
PASS USB_DP has a high-value pull-down to GND (pinout audit F4, chip rev < 3)
PASS HDMI +5V reaches HPDI only through a series resistor [('R67', 50490.0)]: back-feed into an unpowered brid
ge at most 0.10 mA (limit 0.5 mA ASSUMED)
PASS CONFIG_KVM_ATX_PWR_GPIO through 465 ohm: IF 2.47 mA, opto sinks 4.44 mA at CTR min; header pull-up curren
t ASSUMED under 1 mA (ATX front-panel switch inputs are pulled up through kilo-ohms; motherboard specs vary)
PASS CONFIG_KVM_ATX_RST_GPIO through 465 ohm: IF 2.47 mA, opto sinks 4.44 mA at CTR min; header pull-up curren
t ASSUMED under 1 mA (ATX front-panel switch inputs are pulled up through kilo-ohms; motherboard specs vary)
PASS routed tracks were read back from the saved board (1506 segments on 108 nets)
pair            P mm    N mm   skew   vias P/N
HDMI D2         8.94    8.93   0.00   0/0
HDMI D1        10.75   11.08   0.33   0/0
HDMI D0        13.37   13.37   0.00   0/0
HDMI C         15.43   15.52   0.10   0/0
CSI D0         10.10   10.10   0.00   1/1
CSI D1         10.06   10.15   0.09   1/1
CSI C          10.16   10.25   0.09   1/1
USB             9.67    9.50   0.17   3/3
PASS every high speed line has routed length, shortest 8.93 mm
PASS worst P/N length difference 0.33 mm is within 3.0 mm (about 2 ps; one bit at 972 Mbit/s is 1029 ps)
PASS CSI within each pair: worst 0.091 mm against 0.15 mm (Espressif 10 mil = 0.254 mm)
```

```
PASS CSI between pairs: CSI D0 10.10, CSI D1 10.11, CSI C 10.20 mm, spread 0.099 mm against 0.762 mm (Espressi
f 30 mil); about 1 ps
PASS CSI between pairs within the project aim of 0.5 mm: spread 0.099 mm
PASS filled GND on In1 and In4 read back from the saved board (zones refilled)
PASS CSI, TMDS and USB lines have GND under them (In1 under F.Cu, In4 under B.Cu, line +0.03 mm each side)
PASS differential pairs: gap to the partner keeps Zdiff within +-10% on all 16 lines (CSI/TMDS 100, USB 90 oh
m)
WARN reference plane within 0.1 mm of the pair lines: 9 spots at foreign-via antipad edges, 0.0607 mm2 ['CK_
N', 'CK_P', 'CSI_CK_N', 'CSI_CK_P', 'D0_N', 'D1_N', 'USB_DN']
PASS controlled-impedance nets use only the width impedance_r7.py solved for (TMDS/CSI 0.127, USB HS 0.18, LNA
_OUT 0.15)
PASS no via hole touches a BGA ball pad, 318 vias checked
PASS polarized capacitors: + (pin 1) on the supply side, - (pin 2) on GND
PASS no via drill inside a pasted pad, 524 pads checked
PASS every pin level could be bounded (resistor networks small enough to solve at the tolerance corners)
PASS no net has only one pin
PASS every part number decodes to its stated value
PASS every fitted part is stocked at JLC
PASS MASKCHECK mask bridge minimum 0.1088 mm, 0 under 0.1, 3 under 0.13 (black/white only); opening to other-n
et conductor minimum 0.1007 mm, 0 under 0.09; 0 openings merged across nets; openings {'F': {'openings': 213,
'copper': 977, 'not_on_a_pad': 0}, 'B': {'openings': 360, 'copper': 1491, 'not_on_a_pad': 0}}
PASS HOLECHECK 0 under limit; VIA outer min 0.2 (limit 0.2), VIA inner min 0.2012 (limit 0.2), NPTH outer min
none near (limit 0.2), NPTH inner min none near (limit 0.2), PTH outer min 0.3017 (limit 0.28), PTH inner min
0.32 (limit 0.3); holes {'VIA': 318, 'PTH': 16, 'NPTH': 0} on 6 copper layers
PASS component hole to any hole at least 0.45 mm edge to edge (JLC pad hole-to-hole, applied to vias too); und
er it: [] (gap, x, y, component hole, other hole)
PASS BODYCHECK 129 parts, 4908 same-side pairs, 0 overlaps, 0 worst-case warnings, smallest nominal gap 0.350
mm
BOARD sha256 9233a26ad1a1a5a4264edf774e52bf83be37c0ba679b807c393bb46f27976c97
WARNINGS 2
RESULT ALL PASS
```

附錄 5：MUTATION_REPORT.txt

```
mutation run on D:\git\p4-kvm-box\v2_r7f\out
ok  | verify   | control | control passes |
ok  | verify   | U6.5 TCA9517A EN back on 3V3 (the r6 bug) | caught | FAIL  U6.5    3V3        static level of 3V3 3.185-3.440 V against VIH 3.850 V  [TI TCA9517A SCPS245E (rev. Oct 2025) p5 5.3 VIH EN 0.7 x VCCB, max 5.5] / FAIL
ok  | verify   | R50 and R51 swapped (3V3 divider) | caught | FAIL  U1.85   3V3        supply 3.0-3.6 V: 3V3 0.7139-0.7519 V  [Waveshare ESP32-P4-Module wiki: 3.3 V supply; 3.0-3.6 is the P4 VDDPST range, ESP32-P4 datashee
ok  | verify   | R51 33.2k -> 30k (3V3 rises to 3.7 V) | caught | FAIL  U1.85   3V3        supply 3.0-3.6 V: 3V3 3.4624-3.7414 V  [Waveshare ESP32-P4-Module wiki: 3.3 V supply; 3.0-3.6 is the P4 VDDPST range, ESP32-P4 datashee
ok  | verify   | R62 and R63 swapped (1.2 V divider) | caught | FAIL  U7.B2   AVDD12     supply 1.15-1.25 V: AVDD12 69.7247-72.8883 V (after AVDD12 drop 21.5 mV)  [Toshiba TC358743XBG datasheet 2014-05-16 p15 5.2 AVDD12] /
ok  | verify   | R61-R63 at 5% instead of 0.1% | caught | FAIL  U7.B2   AVDD12     supply 1.15-1.25 V: AVDD12 1.1059-1.3017 V (after AVDD12 drop 21.5 mV)  [Toshiba TC358743XBG datasheet 2014-05-16 p15 5.2 AVDD12] /   F
ok  | verify   | HPDO tied to CSI clock+ (two outputs on one net) | caught | FAIL no net has two push-pull or power outputs (pin types from data/pin_types.json) {'CSI_CK_P': ['U7.C8 output', 'U7.H1 output']} / FAIL HDMI connector to TC35
ok  | verify   | Y1 load cap C30 47 pF fitted | caught | FAIL  Y1.3    OSC_OUT    output load 53.6 pF (fitted [('C30', 47.0)] + 6.6 trace/pad/input estimate) against 15 pF  [YXC YSO110TR (LCSC C725995) datasheet, data
ok  | verify   | HPDI straight onto HDMI +5V | caught | FAIL HDMI +5V reaches HPDI only through a series resistor []: back-feed into an unpowered bridge at most 5300.00 mA (limit 0.5 mA ASSUMED) / FAIL no net has onl
ok  | verify   | VDDIO2 on its own unpowered net | caught | FAIL  U7.H7   VDDIO2_X   supply pin on VDDIO2_X, which has no derived supply (unpowered)  [Toshiba TC358743XBG datasheet 2014-05-16 p15 5.2 VDDIO2] /   FAIL  U7
ok  | verify   | TEST ball tied to 3V3 | caught | FAIL  U7.E5   3V3        must be low: 3V3 3.440 V against VIL 0.955 V  [Toshiba TC358743XBG datasheet 2014-05-16 p10 TEST "0: Normal mode", p16 VIL 0.3 VDDIO] /
ok  | verify   | REXT 2.2k instead of 2k | caught | FAIL  U7.A1   TC_REXT    2000 ohm +-1% to 3V3: [('R20', 2178, 2222)]  [Toshiba TC358743XBG datasheet 2014-05-16 p11 REXT "connect to AVDD33 with a 2K ohm resist
ok  | verify   | VBUS divider R10 100k (GPIO3 below VIH) | caught | FAIL  U1.18   HOST_VBUS_DIV static level of HOST_VBUS_DIV 2.349-2.780 V against VIH 2.580 V  [Espressif ESP32-P4 datasheet v0.7 p89 VIH 0.75 VDD, max VDD+0.3] /
ok  | verify   | HDMI D0+ and D0- swapped at the bridge | caught | FAIL HDMI connector to TC358743 by function: TMDS, CEC, DDC via TCA9517A B to A side, HPD from HPDO through a resistor [('D0+', 'HDMID0P'), ('D0-', 'HDMID0N')]
ok  | verify   | CSI D0+ and D1+ swapped at the bridge | caught | FAIL CSI-2: 2 lanes and clock, module pin to TC358743 ball by name, P to P and N to N [('CSI_DATAP0', 'CSID0P'), ('CSI_DATAP1', 'CSID1P')]
ok  | verify   | host VBUS joined to box 5V | caught | FAIL  U1.18   HOST_VBUS_DIV static level of HOST_VBUS_DIV 0.000-0.008 V against VIH 2.580 V  [Espressif ESP32-P4 datasheet v0.7 p89 VIH 0.75 VDD, max VDD+0.3] /
ok  | verify   | R67 not fitted (HPDI floating) | caught | FAIL  U7.H2   HPDI       static level of HPDI unknown against VIH 2.408 V  [Toshiba TC358743XBG datasheet 2014-05-16 p10 HPDI on VDDIO1, p11 Note1 5 V tolerant,
ok  | verify   | U2 EN tied to GND | caught | FAIL  U2.1    GND        static level of GND 0.000-0.000 V against VIH 1.500 V  [Silergy AN_SY8089/A Rev 0.9A p4 VENH 1.5, p2 EN abs VIN+0.6, "do not float"] /
ok  | verify   | U4 TLV73325 replaced by TLV73312 (C2871228) | caught | FAIL every part on the board has its pins in data/pin_types.json ['U4'] /   FAIL  U7.B5   AVDD25     supply pin on AVDD25, which has no derived supply (unpowere
ok  | verify   | I2C SDA moved from GPIO7 to GPIO6 | caught | FAIL firmware sdkconfig.v2r7_rev3: I2C 7/8, bridge reset 23, LCD 9-13, SD 39-45, ATX 2/4/5 each reach the vendor pin they drive [('CONFIG_KVM_I2C_SDA_GPIO', '7'
ok  | verify   | LCD DIN and CLK swapped at J6 | caught | FAIL firmware sdkconfig.v2r7_rev3: I2C 7/8, bridge reset 23, LCD 9-13, SD 39-45, ATX 2/4/5 each reach the vendor pin they drive [('CONFIG_KVM_DISP_MOSI_GPIO', '
ok  | verify   | R41 fitted as well as R42 (3V3 onto VO4) | caught | FAIL SD switch source fed from the module 3.3 V rail, not from ESP_LDO_VO4 (reached through links: ['3V3', 'LDO_VO4', 'OSC_VDD', 'SD_SRC', 'SW'])
ok  | verify   | Q1 gate pull-down R40 not fitted | caught | FAIL  Q1.1    SD_PWR_N   |VGS| at most 3.44 V against 12 V; gate level with the GPIO released: None V (must be defined)  [AOS AO3401A Rev 3.1 p1 VGS +-12 V; af
```

```
ok  | verify   | DDC B-side pull-up R27 to 3V3 instead of 5V | caught | FAIL  U6.7    DDC_SCL    pull-up from
DDC_SCL to 5V (pin 8): []  [TI TCA9517A SCPS245E (rev. Oct 2025) p4] / FAIL electrical rule table: 69 rules fr
om data/pin_
ok  | verify   | USB D+ and D- swapped at the host connector | caught | FAIL USB-C: A/B sides joined, Rd 5.1k
+-20% on each CC, host port on USB_DP/DM, power port on GPIO25/24 (Waveshare alt names) ['host port to USB_DP/
DM']
ok  | verify   | CC pull-down R1 10k instead of 5.1k | caught | FAIL USB-C: A/B sides joined, Rd 5.1k +-20% on
each CC, host port on USB_DP/DM, power port on GPIO25/24 (Waveshare alt names) [('J3', 'A5', 'Rd', [('R1', 990
0.0
ok  | verify   | ESD channel taken off D1- | caught | FAIL every TMDS and USB data line has a TPD4E05U06 chann
el on it ['J2.6']
ok  | verify   | TC358743 VSS ball A2 left open | caught | FAIL every ground pin named in the vendor tables is
on board ground (net of J5 GND) [('U7.A2', 'VSS', 'unconnected-(U7-PadA2)')]
ok  | verify   | VDDC bead FB5 not fitted | caught | FAIL  U7.C3   VDDC       supply pin on VDDC, which has no
derived supply (unpowered)  [Toshiba TC358743XBG datasheet 2014-05-16 p15 5.2 VDDC] /   FAIL  U7.F5
ok  | verify   | ATX PWR opto cathode lifted off GND | caught | FAIL firmware sdkconfig.v2r7_rev3: I2C 7/8, br
idge reset 23, LCD 9-13, SD 39-45, ATX 2/4/5 each reach the vendor pin they drive [('CONFIG_KVM_ATX_PWR_GPIO',
'4'
ok  | verify   | board changed, design.json not regenerated | caught | FAIL design.json and the board agree pi
n for pin (557 pads) ['R11.1'] /   FAIL  U1.87   ESP_EN     pulled high: ESP_EN static None V, VIH_nRST 0.75 x
VDD = 2.58
ok  | verify   | 5 mm stub added to CK_P (skew budget) | caught | FAIL worst P/N length difference 4.90 mm is
within 3.0 mm (about 33 ps; one bit at 972 Mbit/s is 1029 ps) / FAIL CSI, TMDS and USB lines have GND under th
em (In
ok  | verify   | one D1_P segment widened 0.127 -> 0.2 mm | caught | FAIL differential pairs: gap to the partn
er keeps Zdiff within +-10% on all 16 lines (CSI/TMDS 100, USB 90 ohm) [('D1_P', 4, (-4.73, -4.46, 0.162, 'wid
th 0.2 no
ok  | verify   | CSI_CK_P 0.3 mm longer than CSI_CK_N (Espressif 10 mil) | caught | FAIL CSI within each pair:
worst 0.209 mm against 0.15 mm (Espressif 10 mil = 0.254 mm) / FAIL differential pairs: gap to the partner kee
ps Zdiff within +-10% o
ok  | verify   | CSI D0 pair 0.9 mm longer than the others (Espressif 30 mil) | caught | FAIL CSI between pair
s: CSI D0 11.00, CSI D1 10.11, CSI C 10.20 mm, spread 0.898 mm against 0.762 mm (Espressif 30 mil); about 6 ps
ok  | verify   | In1 ground fill missing under the CSI lines | caught | FAIL filled GND on In1 and In4 read ba
ck from the saved board (zones refilled) / FAIL CSI, TMDS and USB lines have GND under them (In1 under F.Cu, I
n4 under B.C
ok  | verify   | TC358743 IR (B4) tied to 3V3 instead of GND | caught | FAIL  U7.B4   3V3        must be low:
3V3 3.440 V against VIL 0.955 V  [Toshiba TC358743XBG datasheet 2014-05-16 p10 IR, Sch input with no pull and
no "when not
ok  | erc      | control | control passes |
ok  | erc      | HPDO output tied to CSI clock+ output | caught | ERC error pin_to_pin: Symbol U7 接腳 C8 [C8,
Output, Line] || Symbol U7 接腳 H1 [H1, Output, Line] / ERC warning single_global_label: 全域標籤 'TC_HPDO'
ok  | erc      | VDDIO2 on its own unpowered net | caught | ERC error power_pin_not_driven: Symbol U7 接腳 H7
[H7, Power input, Line] / ERC warning single_global_label: 全域標籤 'VDDIO2_X'
ok  | erc      | U4 LDO output onto the 3V3 buck rail | caught | ERC error power_pin_not_driven: Symbol U7 接腳
B5 [B5, Power input, Line] / ERC error pin_to_pin: Symbol #FLG01 接腳 1 [pwr, Power output, Line] || Symbol U4
接腳 5 [
ok  | erc      | L1 not fitted (3V3 has no source) | caught | ERC error power_pin_not_driven: Symbol U1 接腳 84
[84, Power input, Line] / ERC error power_pin_not_driven: Symbol Y1 接腳 4 [4, Power input, Line]
ok  | erc      | TC358743 GND ball on a floating net | caught | ERC error power_pin_not_driven: Symbol U7 接腳
A2 [A2, Power input, Line] / ERC warning single_global_label: 全域標籤 'GND_X'
ok  | erc      | REFCLK input left open | caught | U7.H5 (REFCLK, input) is open and pin_types.json open_ok gi
ves no reason
ok  | erc      | Y1 OE input left open | caught | Y1.1 (OE, input) is open and pin_types.json open_ok gives no
reason
ok  | erc      | schematic written the r7b way (every pin passive) | caught | schematic pin types differ from
data/pin_types.json (553 in file, 553 expected): ['U1.1 file passive data power_in', 'U1.3 file passive data p
ower_in', 'U1.4 fi
ok  | erc      | TC358743 IR (B4) left open again | caught | U7.B4 (IR, input) is open and pin_types.json open
_ok gives no reason
ok  | drc      | control | control passes |
```

```
ok  | drc      | new courtyard overlap C32 + C33 (accepted type, new instance) | caught | warning courtyards_o
verlap C33 + R43 (not in data/drc_accepted.json) / warning courtyards_overlap C32 + C33 (not in data/drc_accep
ted.json)
ok  | drc      | new starved thermal at C10.2 (accepted type, new instance; injected into the DRC result) | ca
ught | warning starved_thermal C10.2 + zone GND B.Cu (not in data/drc_accepted.json)
ok  | drc      | LCD_DIN track across D2_P | caught | error clearance @-2.85,-6.19 + U8.4 / error tracks_cross
ing @-2.85,-6.19 + @-3.25,-7.25
ok  | drc      | track 0.095 mm from D2_P (netclass clearance 0.1) | caught | error clearance @-3.46,-6.49 + U
8.6 / error clearance @-3.25,-7.25 + @-3.46,-6.49
ok  | drc      | second via, holes 0.15 mm apart | caught | error hole_to_hole zone 3V3 F.Cu + zone 3V3 F.Cu
ok  | drc      | track 0.05 mm from the board edge | caught | error copper_edge_clearance @-18.50,-20.00 + @0.
00,-19.90 / unconnected  @0.00,-19.90 + zone GND F.Cu
ok  | drc      | one LCD_CS segment deleted (open net) | caught | unconnected  @10.31,12.80 + @11.90,12.80
ok  | drc      | GND via 0.23 mm (hole to hole) from the J5 shell slot | caught | error hole_to_hole J5 + zone
GND F.Cu
ok  | mask     | control | control passes | mask bridge minimum 0.1088 mm, 0 under 0.1, 3 under 0.13 (black/wh
ite only); opening to other-net conductor minimum 0.1007 mm, 0 under 0.09; 0 openings merged a
ok  | mask     | mask expansion 0.05 mm on every pad | caught | mask bridge minimum 0.0088 mm, 37 under 0.1, 9
1 under 0.13 (black/white only); opening to other-net conductor minimum 0.0507 mm, 661 under 0.09; 0 openings
merg
ok  | hole     | control | control passes | 0 under limit; VIA outer min 0.2 (limit 0.2), VIA inner min 0.2012
(limit 0.2), NPTH outer min none near (limit 0.2), NPTH inner min none near (limit 0.2), PTH
ok  | hole     | GND via drilled 0.15 mm from a D2_P track | caught | 8 under limit; VIA outer min -0.1 (limit
0.2), VIA inner min -0.1 (limit 0.2), NPTH outer min none near (limit 0.2), NPTH inner min none near (limit 0.
2), PTH o
ok  | hole     | GND via 0.23 mm (hole to hole) from the J5 shell slot, verify.py | caught | FAIL HOLECHECK 1
under limit; VIA outer min 0.2 (limit 0.2), VIA inner min -0.1 (limit 0.2), NPTH outer min none near (limit 0.
2), NPTH inner min none near (lim
ok  | body     | control | control passes | 129 parts, 4908 same-side pairs, 0 overlaps, 0 worst-case warning
s, smallest nominal gap 0.350 mm
ok  | body     | U6 moved 0.35 mm toward R28 | caught | 129 parts, 4908 same-side pairs, 1 overlaps, 0 worst-c
ase warnings, smallest nominal gap 0.137 mm U6/R28       B  nom  0.137 (area  0.00)  max  0.037  worst -0.
ok  | body     | J6 moved 0.8 mm on the board only (design.json stale) | caught | 128 parts, 4873 same-side pa
irs, 0 overlaps, 0 worst-case warnings, smallest nominal gap 0.350 mm PROBLEMS J6: design.json and the board d
isagree on position
ok  | body     | C20 moved 0.025 mm back toward C19 (margin 0.05) | caught | 129 parts, 4908 same-side pairs,
1 overlaps, 0 worst-case warnings, smallest nominal gap 0.350 mm C19/C20      B  nom  0.375 (area  0.00)  max
0.225  worst  0.
ok  | hotplug  | control | control passes | damping) while the eFuse is off /   worst VBUS_IN peak 10.258 V at
Vs 5.25 L 1.5 uH R 30 mOhm C 1.10 uF; 72 cases (limit 18 V, IN abs max 20 V)
ok  | hotplug  | R72 (EN divider bottom) not fitted: EN sees the VBUS_IN ring | caught | ) while the eFuse is
off /   worst VBUS_IN peak 10.258 V at Vs 5.25 L 1.5 uH R 30 mOhm C 1.10 uF; 72 cases (limit 18 V, IN abs max
20 V)
ok  | hotplug  | R71 and R72 swapped (EN at 68 % of VBUS_IN) | caught | ) while the eFuse is off /   worst VBU
S_IN peak 10.258 V at Vs 5.25 L 1.5 uH R 30 mOhm C 1.10 uF; 72 cases (limit 18 V, IN abs max 20 V)
ok  | imp      | control | control passes | 102.8 /   er 3.8: 101.8
ok  | imp      | outer prepreg 0.0994 -> 0.12 mm in the board stackup | caught | v2_r7f\tools\r7\impedance_r7.
py", line 30, in <module> /     assert (h_outer, er_outer, t_outer) == (S.H_PP, S.ER_PP, S.T_OUT) == (L['diele
ctric 5'][1], L['diel
ok  | fabpack  | verify.out reports a failure | caught | FAIL verify ALL PASS / FAIL verify ran the mask check
ok  | fabpack  | erc_check.out reports a failure | caught | FAIL ERC check (typed pins) passed / FAIL 5V hot-p
lug simulation OK on this design
ok  | fabpack  | schematic edited after erc_check ran | caught | FAIL ERC check ran on this exact schematic /
FAIL 5V hot-plug simulation OK on this design
ok  | fabpack  | hotplug.out reports a peak over 5.8 V | caught | FAIL 5V hot-plug simulation OK on this desig
n / REFUSED: nothing written
ok  | fabpack  | hotplug.out (eFuse form) reports EN over its limit | caught | FAIL 5V hot-plug simulation OK
on this design / REFUSED: nothing written
ok  | r7e      | control (verify after a zone refill) | control passes |
ok  | r7e      | 3V3 via 0.4 mm beside HDMI CK_N (TMDS plane gap, REVIEW injection 6) | caught | FAIL CSI, TMD
```

S and USB lines have GND under them (In1 under F.Cu, In4 under B.Cu, line +0.03 mm each side) [('CK_N', 'B.C
u', 0.0019)]
ok | r7e | 3V3 via 0.4 mm beside CSI_CK_N on F.Cu (CSI plane gap, injection 5) | caught | FAIL CSI, TMDS
and USB lines have GND under them (In1 under F.Cu, In4 under B.Cu, line +0.03 mm each side) [('CSI_CK_N', 'F.C
u', 0.0019)]
ok | r7e | control (r7d_csi_ref.py exit code) | control passes | exit 0 RESULT PASS
ok | r7e | r7d_csi_ref.py on the CSI plane-gap via: must exit non-zero | caught | exit 1 FAIL CSI_CK_N
F.Cu gap 0.0019 mm2 at (7.316, -3.367); other-net vias within 0.8 mm: [(0.31, 'CSI_D1_N')] / RESULT FAIL
ok | r7e | control (pairgap) | control passes |
ok | r7e | TMDS D2_N segment 0.12 mm closer to D2_P (gap 0.198 -> 0.078) | caught | FAIL D2_P: impedance
outside 100 ohm +-10% at 67 samples (gap to D2_N), first [(-2.97, -5.0, 0.108, 88.0), (-3.12, -4.99, 0.097, 8
6.3), (-2.32, -5.0, 0.108, 88.
ok | r7e | USB_DN segment 0.2 mm closer to USB_DP (gap 0.32 -> 0.12, 78 ohm) | caught | FAIL USB_DP: imp
edance outside 90 ohm +-10% at 86 samples (gap to USB_DN), first [(8.97, -10.55, 0.12, 77.8), (8.92, -10.55,
0.12, 77.8), (8.88, -10.55, 0.12, 7
ok | r7e | GND via put back in the middle of C21.2 (via in pad, REVIEW m2) | caught | FAIL no via drill
inside a pasted pad, 524 pads checked [('C21.2', 'GND', (1.65, -8.43))]
ok | r7e | C69 reversed: + on GND, - on 5V_DAMP (REVIEW injection 3), verify | caught | FAIL polarized c
apacitors: + (pin 1) on the supply side, - (pin 2) on GND [('C69', '+ GND', '- 5V_DAMP')]
ok | r7e | control (CPL written as fab_pack does, cpl_pin1) | control passes |
ok | r7e | C69 reversed, cpl_pin1 | caught | FAIL C69: polarized capacitor pin 1 (+) on GND, pin 2 on 5V
_DAMP: + must reach a supply (['ATX_LED', 'EF_EN', 'EF_ILM', 'FB_1V2', 'FB_3V3', 'GND', 'HDMI_HPD', '
ok | r7e | U2 CPL angle 180 degrees off (SY8089 on the bottom) | caught | FAIL U2: FAIL: CPL 90, KiCad 0
+ verified offset 270 = 270
ok | r7e | U4 footprint pin 1 moved (pads 1 and 3 swapped) | caught | FAIL SOT-23-5: pin 1 in the footpr
int frame differs between [('U4', (0.95, 1.138)), ('U2', (-0.95, 1.137)), ('U15', (-0.95, 1.138))] (rotation c
onvention or foo
ok | case | control | control passes |
ok | case | J6 moved 0.8 mm / 3.6 mm toward the M2 screw | caught | FAIL M2 shank clears the J6 envelope
[-0.36 mm (x gap alone -0.78 mm)] / FAIL plate clears every top-side part (J6 through its cut-out) [[('J6',
2.5)]]
ok | r7f | control (verify) | control passes |
ok | r7f | J5 VBUS wired straight to 5V (eFuse bypassed) | caught | FAIL U16.3 VBUS_IN supply pin
on VBUS_IN, which has no derived supply (unpowered) [TI TPS2595xx SLVSE57C (Apr 2018), datasheets_r7f/tps259
5.pdf p5 7.3 VI
ok | r7f | D1 (CEC ESD) not fitted | caught | FAIL r7f: HDMI CEC (HDMI_CEC) has a TPD1E10B06 clamp to GN
D []
ok | r7f | R80 (LCD DIN series) 330 ohm instead of 33 | caught | FAIL firmware sdkconfig.v2r7_rev3: I2C
7/8, bridge reset 23, LCD 9-13, SD 39-45, ATX 2/4/5 each reach the vendor pin they drive [('CONFIG_KVM_DISP_MO
SI_GPIO', '
ok | r7f | R76 (SD D0 series) 220 ohm instead of 22 | caught | FAIL firmware sdkconfig.v2r7_rev3: I2C 7/
8, bridge reset 23, LCD 9-13, SD 39-45, ATX 2/4/5 each reach the vendor pin they drive [('CONFIG_KVM_SD_D0_GPI
O', '39',
ok | r7f | 0 ohm 5V(pin 1) -> VBUS_IN(pin 2) across the eFuse | caught | FAIL r7f: nothing bypasses the
eFuse U16 between VBUS_IN and 5V (any part, either pin order) ['C70']
ok | r7f | 0 ohm VBUS_IN(pin 1) -> 5V(pin 2) across the eFuse | caught | FAIL r7f: J5 VBUS reaches 5V on
ly through the eFuse ['U16'] (IN VBUS_IN, OUT 5V) / FAIL r7f: nothing bypasses the eFuse U16 between VBUS_IN a
nd 5V (any part, ei
ok | r7f | R73 (ILIM) 10k instead of 1k | caught | FAIL r7f: U16 ILM (EF_ILM) has one resistor to GND wi
thin 487-5000 ohm (SLVSE57C 7.3), current limit 0.24-0.24 A (Eq. 4) [('R73', '1')] [('R73', 'EF_ILM', 'GND'
ok | r7f | R73 (ILIM) not fitted | caught | FAIL r7f: U16 ILM (EF_ILM) has one resistor to GND within 48
7-5000 ohm (SLVSE57C 7.3), current limit n/a (Eq. 4) [] [] / FAIL r7f: U16 ILIM at its low end 0.00
ok | r7f | R73 (ILIM) 2.2k (C25879, ILIM about 0.95 A, below the full load) | caught | FAIL r7f: U16 ILI
M at its low end 0.87 A carries the full load 1.51 A x 1.1 = 1.66 A (Eq. 4 at R max, ILIM -7.5 %)
ok | r7f | C71 (dVdt) 2.2 uF (C385032, ramp about 260 ms) | caught | FAIL r7f: U16 dVdt ramp to 5 V 288.
4 ms at CdVdt +10 % within the 30 ms start-up budget (Eq. 3; CdVdt 2200.0 nF)
ok | r7f | U16 EP paste windows removed | caught | FAIL r7f: U16 EP paste: two B.Paste windows on the E
P, 0.0 % of it (80-95 %, TI DSG0008A stencil), EP pad itself without paste (0 windows, inside True)
ok | r7f | H1 copper ring back to 3.1 mm (0.218 mm to the board edge) | caught | FAIL r7f: mounting hole
copper ring <= 2.9 mm or copper to the board edge >= 0.3 mm (REVIEW_r7f fix kept) [('H1.1', 3.1, 0.218), ('H2.

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1', 2.9, 0.318), ('H3.1',
ok  | r7f      | C71 and R73 swapped (dVdt on 1k, ILM on 100 nF) | caught | FAIL r7f: U16 ILM (EF_ILM) has one resistor to GND within 487-5000 ohm (SLVSE57C 7.3), current limit n/a (Eq. 4) [('C71', '2')] [] / FAIL r7f: U16 ILIM at its l
ok  | r7f      | R71 (EN top) not fitted | caught | FAIL r7f: U16 EN/UVLO divider from VBUS_IN and to GND, UVLO rising None V within 3.0-4.6 V (VUVLO(R) 1.13-1.27 V, SLVSE57C 7.5) [('R72', '2')]
ok  | r7f      | C71 (dVdt) not fitted | caught | FAIL r7f: U16 dVdt (EF_DVDT) has one capacitor to GND of at least 3300 pF (SLVSE57C 7.3) [] / FAIL r7f: U16 dVdt ramp to 5 V None ms at CdVdt +10 % within the 3
ok  | r7f      | U16 IN and OUT swapped (must FAIL by name, not crash) | caught | FAIL  Q1.1    SD_PWR_N   |VGS| at most 0.00 V against 12 V; gate level with the GPIO released: 0.0 V (must be defined)  [AOS AO3401A Rev 3.1 p1 VGS +-12 V; a fl
ok  | r7f      | H1 mounting hole pad off GND | caught | FAIL r7f: every mounting hole is a plated pad on GND (lid and plate ground through the posts) [('H1.1', ''), ('H2.1', 'GND'), ('H3.1', 'GND'), ('H4.1', 'GND')]
ok  | r7f      | control (csi_via) | control passes | RESULT PASS
ok  | r7f      | 3V3 via 0.5 mm from CSI_D1_P | caught | FAIL via ('3V3', 4.841, -4.623, 'CSI_D0_N', 0.057) / RESULT FAIL
ok  | r7f      | VDD_MIPI via back at its r7d place (0.585 mm from CSI_CK_P) | caught | FAIL via ('VDD_MIPI', 8.18, -1.686, 'CSI_CK_P', 0.585) / RESULT FAIL
ok  | r7f      | control (hand_solder_gap) | control passes | RESULT PASS
ok  | r7f      | R80 0.25 mm toward U1 (pad 0.137 mm from U1.24) | caught | FAIL R80.1 is 0.137 mm from U1.24 (limit 0.2) / FAIL R80.2 is 0.137 mm from U1.24 (limit 0.2)
ok  | r7f      | R81 0.08 mm toward R80 (pads 0.08 mm apart) | caught | FAIL R80 to R81 0.07999999999998408 mm (limit 0.15) / RESULT FAIL
ok  | r7f      | R79 (SD_D3, 0402) 0.75 mm toward U1.70/71 (0.14 mm) | caught | FAIL R79.1 is 0.140 mm from U1.70 (limit 0.2) / FAIL R79.2 is 0.140 mm from U1.71 (limit 0.2)
ok  | r7f      | C3 0.5 mm toward U1.79/80 (0.14 mm) | caught | FAIL C3.1 is 0.140 mm from U1.80 (limit 0.2) / FAIL C3.2 is 0.140 mm from U1.79 (limit 0.2)
ok  | r7f      | control (case with plate, vents, gap filler) | control passes |
ok  | r7f      | spreader plate file missing | caught | FAIL  r7f plate, gap filler and case_spec files exist / RESULT FAIL: 72 passed, 1 failed, case files C:\Users\IIIT\AppData\Local\Temp\mutate_muxe9l_r\r7f_118\st
ok  | r7f      | tray material PLA | caught | FAIL  lid is aluminium, black anodized; tray is heat resistant (not PLA)  [lid Al 6061-T6, black anodized (emissivity ~0.85) except the masked contact faces (ba
ok  | r7f      | lid material plastic | caught | FAIL  lid is aluminium, black anodized; tray is heat resistant (not PLA)  [lid PETG, printed; tray ASA (or ABS / PC / PA12; heat deflection above 90 C; not PLA)
ok  | r7f      | post end faces not masked (anodized: no ground contact) | caught | FAIL  every lid contact face that carries ground is masked from the anodize  [masked ['lid_rim', 'post_bore', 'post_shoulder']; missing ['post_end']] / RESULT F
ok  | r7f      | ground resistance acceptance test dropped | caught | FAIL  lid-to-GND and plate-to-GND < 1 ohm are assembly acceptance tests  [[]] / RESULT FAIL: 98 passed, 1 failed, case files C:\Users\IIIT\AppData\Local\Temp\mu
ok  | r7f      | fan spec says powered from the board | caught | FAIL  the fan is powered from its own USB supply, not from the board  [30 x 30 mm 5 V fan powered from the board 5V] / RESULT FAIL: 98 passed, 1 failed, case fi
ok  | r7f      | fan duct file missing | caught | FAIL  fan duct files exist / RESULT FAIL: 91 passed, 1 failed, case files C:\Users\IIIT\AppData\Local\Temp\mutate_muxe9l_r\r7f_124\stl
ok  | r7f      | H2 mounting hole pad off GND (case grounding check) | caught | FAIL  four mounting holes are plated and on GND  [[(18, 23, 'thru_hole', 2.9, 'GND'), (-18, -18, 'thru_hole', 2.9, None), (18, -18, 'thru_hole', 2.9, 'GND'), (-

126 runs, 0 bad
MUTATION ALL CAUGHT
```

附錄 6：case_check.txt

```
===== r7f board 40 x 45 =====
board p4_kvm_mini.kicad_pcb sha256 9233a26ad1a1a5a4, solids 590, outline 40.00 x 45.00, thru-plated 16
PASS  Gerber pad positions agree with STEP pad solids  [misses F 0 B 0, of 213 + 360]
PASS  board top is +z in the STEP
PASS  four screw holes found in the board  [[(18.0, 23.0, 2.2), (-18.0, 23.0, 2.2), (18.0, -18.0, 2.2), (-18.
0, -18.0, 2.2)]]
real connector centres in the case frame: J2 (8.00, -12.86) nearest wall -y, J3 (-8.75, -14.61) nearest wall -
y, J5 (-14.61, 2.50) nearest wall -x, J7 (19.01, 4.20) nearest wall +x, J4 (3.70, 14.90) nearest wall +y, J6
(-12.10, 13.30) nearest wall -x
standing outside the I/O face (-y) +x is on your right: J3 LEFT, J2 RIGHT
PASS  real J3 is left of real J2, seen from outside the I/O face
PASS  connector sides as the z windows below assume  [J3 J5 J6 top, J2 J7 J4 bottom]
PASS  tray STL and STEP are the same part  [volume differs 0.01%]
PASS  lid STL and STEP are the same part  [volume differs 0.02%]
PASS  J2 on the -y wall is the wall nearest its real position  [nearest -y]
PASS  J2 opening in -y wall clears the HDMI shell 15 x 6  [0.00 mm2 of wall in the way at u 8.00]
PASS  J2 opening centred on the connector  [offset +0.00, opening u -0.10..16.10 z 2.10..8.90]
PASS  J2 plug overmould 19.5 x 11.0 fits its pocket  [0.00 mm2 in the way]
PASS  J3 on the -y wall is the wall nearest its real position  [nearest -y]
PASS  J3 opening in -y wall clears the USB-C shell 8.94 x 3.2  [0.00 mm2 of wall in the way at u -8.75]
PASS  J3 opening centred on the connector  [offset +0.00, opening u -13.70..-3.80 z 9.25..13.35]
PASS  J3 plug overmould 12.0 x 8.0 fits its pocket  [0.00 mm2 in the way]
PASS  J5 on the -x wall is the wall nearest its real position  [nearest -x]
PASS  J5 opening in -x wall clears the USB-C shell 8.94 x 3.2  [0.00 mm2 of wall in the way at u 2.50]
PASS  J5 opening centred on the connector  [offset +0.00, opening u -2.45..7.45 z 9.25..13.35]
PASS  J5 plug overmould 12.0 x 8.0 fits its pocket  [0.00 mm2 in the way]
PASS  J7 on the +x wall is the wall nearest its real position  [nearest +x]
PASS  J7 opening in +x wall clears the card 11 wide at the shell  [0.00 mm2 of wall in the way at u 4.20]
PASS  J7 opening centred on the connector  [offset -0.10, opening u -2.20..10.40 z 6.30..8.45]
PASS  J4 on the +y wall is the wall nearest its real position  [nearest +y]
PASS  J4 opening in +y wall clears the SHR-06V plug 8 wide  [0.00 mm2 of wall in the way at u 3.70]
PASS  J4 opening centred on the connector  [offset -0.00, opening u -0.90..8.30 z 3.90..8.45]
PASS  HDMI and host USB plugs fit side by side  [1.00 mm between overmoulds]
PASS  label HDMI is cut in the -y wall beside J2  [text u 4.6..9.4 z 14.5..15.6, connector u 8.00]
PASS  label MF KVM is cut in the -y wall beside J2  [text u 1.2..12.8 z 15.9..18.3, connector u 8.00]
PASS  label HOST is cut in the -y wall beside J3  [text u -11.3..-6.1 z 16.4..18.1, connector u -8.75]
PASS  label 5V is cut in the -x wall beside J5  [text u 1.2..3.7 z 16.4..18.1, connector u 2.50]
PASS  label TF is cut in the +x wall beside J7  [text u 2.8..5.3 z 10.2..11.9, connector u 4.20]
PASS  label ATX is cut in the +y wall beside J4  [text u 1.7..5.7 z 9.7..11.4, connector u 3.70]
vents -y 0, +y 4, -x 9, +x 0
PASS  no vents on the +x (microSD) wall or the I/O wall
PASS  -x wall vent beside U7  [U7 real at (-3.00, -4.00), nearest slot 0.70 mm along the wall]
PASS  -x wall vent beside U2  [U2 real at (-14.00, 10.08), nearest slot 1.38 mm along the wall]
PASS  -x wall vent beside L1  [L1 real at (-13.60, 13.83), nearest slot 0.87 mm along the wall]
PASS  back wall vent behind the buck stage  [0.90 mm from L1 along the wall]
PASS  tray clearance bore at (18, 23) lines up  [d 2.4, offset 0.0]
PASS  tray head recess at (18, 23) takes the M2 head  [d 4.2]
PASS  tray post at (18, 23) seats the PCB at z 8.5  [10.7 of 11.4 mm2]
PASS  lid pilot at (18, 23) lines up at z 16.0  [d 1.6, wall 1.95]
PASS  lid post at (18, 23) is whole down at the PCB  [d 1.6, wall 1.10]
PASS  lid post at (18, 23) meets the PCB top  [6.9 of 6.9 mm2]
PASS  tray clearance bore at (-18, 23) lines up  [d 2.4, offset 0.0]
PASS  tray head recess at (-18, 23) takes the M2 head  [d 4.2]
PASS  tray post at (-18, 23) seats the PCB at z 8.5  [10.7 of 11.4 mm2]
PASS  lid pilot at (-18, 23) lines up at z 16.0  [d 1.6, wall 1.95]
PASS  J6 corner (-18, 23): the thread has no thin wall beside J6  [post starts above the J6 relief]
```

```
PASS  tray clearance bore at (18, -18) lines up  [d 2.4, offset 0.0]
PASS  tray head recess at (18, -18) takes the M2 head  [d 4.2]
PASS  tray post at (18, -18) seats the PCB at z 8.5  [10.7 of 11.4 mm2]
PASS  lid pilot at (18, -18) lines up at z 16.0  [d 1.6, wall 1.95]
PASS  lid post at (18, -18) is whole down at the PCB  [d 1.6, wall 1.10]
PASS  lid post at (18, -18) meets the PCB top  [6.9 of 6.9 mm2]
PASS  tray clearance bore at (-18, -18) lines up  [d 2.4, offset 0.0]
PASS  tray head recess at (-18, -18) takes the M2 head  [d 4.2]
PASS  tray post at (-18, -18) seats the PCB at z 8.5  [10.7 of 11.4 mm2]
PASS  lid pilot at (-18, -18) lines up at z 16.0  [d 1.6, wall 1.95]
PASS  lid post at (-18, -18) is whole down at the PCB  [d 1.6, wall 1.10]
PASS  lid post at (-18, -18) meets the PCB top  [6.9 of 6.9 mm2]
PASS  M2 x 16 tip stops inside the pilot  [seat z 1.8, tip z 17.8, pilot top z 18.4]
PASS  thread engagement in the J6 corner is at least 1.5 x d  [3.5 mm]
PASS  every part envelope holds its real pads after the design-to-case map  [137 parts]
PASS  no part body touches the tray or lid  [none]
PASS  real PCB solid (outline, holes, slots) against tray and lid  [tray 0.000, lid 0.000 mm3]
PASS  PCB z in the case is the stack height  [8.54..9.61]
PASS  tray against lid  [0.000 mm3]
PASS  LCD module (estimate) against lid and tray  [lid 0.000, tray 0.000 mm3]
PASS  LCD clears the tallest top-side part  [LCD bottom 15.40, tallest top 13.95]
J6 mouth faces -x, plug and wire keep-out x -19.87..-17.17 y 8.80..17.80 z 9.7..12.6
PASS  J6 plug and wires clear the lid  [0.00 mm3]
PASS  M2 shank clears the J6 envelope  [3.32 mm (x gap alone 0.02 mm)]
PASS  r7f plate, gap filler and case_spec files exist
PASS  plate STL and STEP are the same part  [volume differs 0.00%]
PASS  plate is 0.8 mm Al at z 13.8..14.6  [z 13.80..14.60, Al 5052 or 6061, 0.8 mm, bare (no anodize, no pain
t); no thermal paste on any contact face (posts, shoulders, rim)]
PASS  plate spans the case outline (edge clamped between the halves)  [44.00 x 49.00]
PASS  plate against tray and lid  [tray 0.000, lid 0.000 mm3]
PASS  plate clears every top-side part (J6 through its cut-out)  [tallest under the plate 13.01, plate bottom
13.80]
PASS  plate clears the LCD (estimate) by 0.5 mm or more  [0.80 mm]
PASS  J6 plug and wires clear the plate  [0.00 mm3]
PASS  gap filler covers exactly the P4 12 x 12 and C6 7 x 7 squares  [footprint 186.8 mm2, wanted 186.8 mm2 (t
he two squares overlap at the module corner, as in the thermal model)]
PASS  gap filler sits on the module shield and reaches the plate  [z 13.01..13.80, inside the module outline T
rue]
PASS  gap filler touches nothing but the shield and the plate
PASS  lid top has 12 vent slots, about 72 mm2  [12 slots, 72.0 mm2]
PASS  vent slots stay outside the LCD seat ring
PASS  lid is aluminium, black anodized; tray is heat resistant (not PLA)  [lid Al 6061-T6, black anodized (emi
ssivity ~0.85) except the masked contact faces (bare Al or conductive chromate conversion); tray ASA (or ABS /
PC / PA12; heat deflection above 90 C; not PLA)]
PASS  four mounting holes are plated and on GND  [[(18, 23, 'thru_hole', 2.9, 'GND'), (-18, -18, 'thru_hole',
2.9, 'GND'), (18, -18, 'thru_hole', 2.9, 'GND'), (-18, 23, 'thru_hole', 2.9, 'GND')]]
PASS  at least three lid posts bear on a plated GND ring (the J6-corner post stops above J6)  [3 of 4]
PASS  the plate is clamped by all four post shoulders (metal contact to the lid)  [4 of 4]
PASS  every lid contact face that carries ground is masked from the anodize  [masked ['lid_rim', 'post_bore',
'post_end', 'post_shoulder']; missing []]
PASS  the plate is bare metal on its contact faces  [Al 5052 or 6061, 0.8 mm, bare (no anodize, no paint); no
thermal paste on any contact face (posts, shoulders, rim)]
PASS  lid-to-GND and plate-to-GND < 1 ohm are assembly acceptance tests  [['lid to board GND (J5 USB-C shell)
< 1 ohm, measured after assembly with a 4-wire or low-range ohmmeter', 'plate to board GND < 1 ohm, measured t
he same way']]
PASS  fan duct files exist
PASS  fan duct against lid, tray, plate  [lid 0.000, tray 0.000 mm3]
PASS  fan duct sits on the lid top  [duct bottom z 20.00, lid top 20.0]
PASS  every lid slot opens into the duct  [12 of 12 slots inside the duct opening]
PASS  the duct stays clear of the LCD window  [0.50 mm]
```

```
PASS  the duct floor closes the part that overhangs the case  [659.3 of 659.3 mm2]
PASS  fan seat: one opening and four screw holes on a 24 mm pitch (30 mm fan)  [1 opening(s), screw holes [(-1
2.0, 20.2), (-12.0, 44.2), (12.0, 20.2), (12.0, 44.2)]]
PASS  the fan is powered from its own USB supply, not from the board  [30 x 30 mm 5 V axial fan, suction, on t
he duct top; powered from its own USB-A/USB-C 5 V supply, NOT from the board (board 5V budget unchanged)]
tray bbox 44.00 x 49.00 x 13.80, lid bbox 44.00 x 49.00 x 10.30
RESULT PASS: 99 passed, 0 failed, case files D:\git\p4-kvm-box\v2_r7f\case\stl
```

附錄 7：hand_solder_gap_r7f.txt

```
board out/p4_kvm_mini.kicad_pcb; pad_to_mask_clearance 0.0
C3.1 (-17.200, 7.780): nearest U1 pad U1.80 0.640 mm, mask web 0.640 mm
C3.2 (-17.200, 6.820): nearest U1 pad U1.79 0.640 mm, mask web 0.640 mm
J2.20 (-15.250, -9.880): nearest U1 pad U1.66 0.988 mm, mask web 0.988 mm
J2.21 (-0.750, -9.880): nearest U1 pad U1.53 0.280 mm, mask web 0.280 mm
J5.A1B12 (11.598, 5.700): nearest U1 pad U1.33 0.623 mm, mask web 0.623 mm
J5.A4B9 (11.598, 4.900): nearest U1 pad U1.34 0.623 mm, mask web 0.623 mm
J5.A5 (11.598, 3.750): nearest U1 pad U1.35 0.623 mm, mask web 0.623 mm
J5.A6 (11.598, 2.750): nearest U1 pad U1.36 0.623 mm, mask web 0.623 mm
J5.A7 (11.598, 2.250): nearest U1 pad U1.37 0.623 mm, mask web 0.623 mm
J5.A8 (11.598, 1.250): nearest U1 pad U1.38 0.623 mm, mask web 0.623 mm
J5.B1A12 (11.598, -0.700): nearest U1 pad U1.39 0.623 mm, mask web 0.623 mm
J5.B4A9 (11.598, 0.100): nearest U1 pad U1.39 0.623 mm, mask web 0.623 mm
J5.B5 (11.598, 0.750): nearest U1 pad U1.38 0.623 mm, mask web 0.623 mm
J5.B6 (11.598, 1.750): nearest U1 pad U1.37 0.623 mm, mask web 0.623 mm
J5.B7 (11.598, 3.250): nearest U1 pad U1.36 0.623 mm, mask web 0.623 mm
J5.B8 (11.598, 4.250): nearest U1 pad U1.35 0.623 mm, mask web 0.623 mm
R79.1 (-17.450, -1.480): nearest U1 pad U1.70 0.890 mm, mask web 0.890 mm
R79.2 (-17.450, -0.520): nearest U1 pad U1.71 0.890 mm, mask web 0.890 mm
R80.1 (10.787, 14.580): nearest U1 pad U1.24 0.387 mm, mask web 0.387 mm
R80.2 (10.787, 15.120): nearest U1 pad U1.24 0.387 mm, mask web 0.387 mm
R81.1 (10.787, 13.580): nearest U1 pad U1.25 0.387 mm, mask web 0.387 mm
R81.2 (10.787, 14.120): nearest U1 pad U1.25 0.387 mm, mask web 0.387 mm
22 top-side pads within 1 mm of a U1 pad checked
R80 to R81 pad to pad 0.160 mm
RESULT PASS
```

附錄 8：csi_via_r7f.txt

```
93 CSI segments, 318 vias; closest non-GND via 0.654 mm ('VDD_MIPI', 8.25, -1.686, 'CSI_CK_P')
RESULT PASS
```

附錄 9：hotplug_report.txt

```
design out/design.json sha256 b8cdf4addbbebc1eb306b24dee0f52bded73918058cb87ee5794b85efaacc381
r7f eFuse U16: J5 rings into VBUS_IN (1.10-1.76 uF effective, no damping) while the eFuse is off
  worst VBUS_IN peak 10.258 V at Vs 5.25 L 1.5 uH R 30 mOhm C 1.10 uF; 72 cases (limit 18 V, IN abs max 20 V)
  EN/UVLO peak 3.280 V through 100.0k/47.0k (limit 6 V, abs max 7 V)
  5V (U2 VIN): the eFuse output is a 12 ms dV/dt ramp, clamped at VCLAMP 5.7 V max (TPS25953x): <= 5.7 V, unde
r 5.8 V
HOTPLUG WORST 10.258 V at VBUS_IN, 3.280 V at EN, U2 VIN <= 5.70 V OK
```

附錄 10：coldstart_r7f.txt

```
Cold plug-in with full load on every rail (r7e analysis, REVIEW_r7d m5). r7f: the U16 eFuse as built is modell
ed in sim_ef().

Time-domain model, 0-60 ms after a USB-C cable from a host port is plugged into J5 with nothing on the board c
harged:
  host:  Vs behind the port's power switch (on-resistance R_sw, current limit I_lim), port bulk C_host (USB 2.
0 7.2.4.1: >= 120 uF)
  cable: R_cable + L_cable (as tools/hotplug.py)
  board 5V: ceramics C60 22u + C64 1u + C32 100n (DC-bias derated) and the damping branch R68 0.15 ohm + C69 3
30 uF (ESR ASSUMED 25 mOhm)
  U2  SY8089 5V->3V3: on when VIN rises above UVLO 2.5 V (max, rising; 0.2 V hysteresis) with EN = VIN above
1.5 V; soft start 1.2 ms
      (typ, the only number given) from 0 to 3.311 V; output limited to VIN - I x (RON_P 110 mOhm + DCR 80 mOh
m); PFET limit 3.5 A (min).
      Input current = output power / 0.90 / VIN.        (SY8089 DS: VIN 2.7-5.5, ILIM >= 3.5 A, VENH 1.5 V, VU
VLO 2.5 V max, TSS 1.2 ms)
  3V3 load, full, from the moment 3V3 exists (resistive below 3.3 V): ESP32-P4 380 mA (Espressif HW Design Gui
delines p10: minimum
      supply 380 mA incl. flash/PSRAM) + ESP32-C6 Wi-Fi TX 354 mA (C6 DS v1.5 p66, 802.11b 21 dBm) + TC358743
AVDD33/VDDIO 142 mA (DS p8
      88 mA at 1080p60, x1.5 ASSUMED: no max published) + LCD 60 mA (ASSUMED: Waveshare 1.28in wiki gives no c
urrent) + microSD 200 mA
      (SD Physical Layer spec, high-speed class, from memory: NOT checked against a document here) + oscillato
r 15 mA + misc 17 mA
      = 1.168 A; plus U15 and U4 input current below.  3V3 capacitance: C1/C2/C10 10 uF, C61/C62/C65 22 uF, 10
0 nF x5 at 3.3 V bias
      (ASSUMED 65 %) + 20 uF inside the module (ASSUMED; the module schematic lists several 10 uF on ESP_3V3).
  U15 SY8089 3V3->1V2: on when 3V3 > 2.5 V; soft start 1.2 ms to 1.2102 V; load 274 mA (TC358743 VDDC+VDD_MIPI
+AVDD12 1080p60 182 mA,
      DS p8, x1.5 ASSUMED), efficiency 0.85 (ASSUMED); V1V2 capacitance 54 uF + 3 x 4.7 uF at 1.2 V (ASSUMED 8
5 %).
  U4  TLV73325 3V3->2V5: follows 3V3 less 0.2 V dropout up to 2.5 V; 19 mA.
  CHIP_PU (ESP_EN): R11 10k from 3V3, C5 1 uF; the P4 starts when it crosses 0.75 x 3V3 (ASSUMED threshold; DS
Table 2-14 asks the rails
      to be stable 50 us before).
Wi-Fi TX at start-up: in the full-load runs the 354 mA is drawn from t = 0 (worst for the start-up). A separat
e run starts from a
settled board with the C6 receiving (80 mA, C6 DS p67) and steps to 354 mA TX; it reports the 3V3 dip with the
converter's loop modelled
as a 50 kHz crossover (ASSUMED) and the 5V droop through the cable and the port limit.
Corners: Vs 4.75 / 5.0 / 5.25 V; cable 0.1 / 0.3 ohm, 1 uH; port: bench supply (no limit), PC port with a 1.0
A limit switch (typical
TPS2051-class, ASSUMED), strict 0.55 A limit (USB 2.0 high-power port at its 500 mA budget). Everything ASSUME
D is listed above;
nothing here is measured.
usage: coldstart.py [--plot out.png] [--report out.txt]

Cold plug-in, full load on every rail (tools/coldstart.py; every input is listed in its docstring)

port                           Vs  Rcab |  Ipk A  at us  Q10ms uC  Qexcess | Vhost min 5V max 5V min>2ms | 3V
3 95% ms 2V5 ms 1V2 ms CHIP_PU ms |  Iss A 3V3 end
--- load FULL: 3V3 1.168 A direct + 1V2 0.274 A ---
bench supply, no limit        4.75  0.10 |  16.39      8     11243      715 |      3.80   4.69       4.56 |
1.17   0.96   2.10      14.50 |   1.05   3.311
bench supply, no limit        4.75  0.30 |  10.08      6     11593      502 |      4.14   4.58       4.33 |
1.23   1.03   2.16      14.56 |   1.11   3.311
```

```
bench supply, no limit        5.00  0.10 |  17.25      8     10811      853 |      4.00   4.94       4.82 |
1.17   0.96   2.09      14.50 |   1.00   3.311
bench supply, no limit        5.00  0.30 |  10.61      6     11099      671 |      4.36   4.83       4.60 |
1.22   1.02   2.15      14.55 |   1.04   3.311
bench supply, no limit        5.25  0.10 |  18.11      8     10435      984 |      4.21   5.19       5.08 |
1.17   0.96   2.10      14.50 |   0.94   3.311
bench supply, no limit        5.25  0.30 |  11.14      6     10671      825 |      4.58   5.08       4.88 |
1.22   1.01   2.14      14.55 |   0.98   3.311
PC port, 1.0 A limit          4.75  0.10 |  16.20      7     10242      241 |      1.61   3.79       2.62 |
1.72   1.52   2.65      never |   1.00   2.335
PC port, 1.0 A limit          4.75  0.30 |  10.05      6     10218      217 |      1.88   3.78       2.62 |
1.74   1.53   2.66      never |   1.00   2.335
PC port, 1.0 A limit          5.00  0.10 |  17.05      8     10272      271 |      1.68   3.79       2.62 |
1.69   1.48   2.62      never |   1.00   2.335
PC port, 1.0 A limit          5.00  0.30 |  10.57      6     10247      247 |      1.95   3.78       2.62 |
1.71   1.50   2.64      never |   1.00   2.335
PC port, 1.0 A limit          5.25  0.10 |  17.90      7     10301      301 |      1.75   3.79       2.62 |
1.66   1.45   2.59      never |   1.00   2.335
PC port, 1.0 A limit          5.25  0.30 |  11.10      6     10277      277 |      2.02   3.78       2.62 |
1.68   1.47   2.60      never |   1.00   2.335
strict port, 0.55 A limit     4.75  0.10 |  16.16      7      5711     -763 |      1.45   2.94       2.30 |
never   never   never      never |   0.65   2.179
strict port, 0.55 A limit     4.75  0.30 |  10.03      6      5701     -336 |      1.62   2.94       2.30 |
never   never   never      never |   0.60   2.228
strict port, 0.55 A limit     5.00  0.10 |  17.01      7      5744    -1072 |      1.52   2.98       2.30 |
never   never   never      never |   0.68   2.325
strict port, 0.55 A limit     5.00  0.30 |  10.55      6      5735     -280 |      1.68   2.94       2.30 |
never   never   never      never |   0.60   2.202
strict port, 0.55 A limit     5.25  0.10 |  17.85      7      5777     -575 |      1.59   3.13       2.30 |
never   never   never      never |   0.64   2.451
strict port, 0.55 A limit     5.25  0.30 |  11.08      6      5771     -227 |      1.75   2.94       2.30 |
never   never   never      never |   0.60   2.181
--- load BOOT: 3V3 0.319 A direct + 1V2 0.010 A ---
bench supply, no limit        4.75  0.10 |  16.39      8      4184     1505 |      3.80   4.71       4.70 |
1.17   0.96   2.10      14.50 |   0.27   3.311
bench supply, no limit        4.75  0.30 |  10.08      6      4179     1469 |      4.14   4.65       4.65 |
1.23   1.03   2.16      14.56 |   0.27   3.311
bench supply, no limit        5.00  0.10 |  17.25      8      4140     1598 |      4.00   4.96       4.95 |
1.17   0.96   2.09      14.50 |   0.25   3.311
bench supply, no limit        5.00  0.30 |  10.61      6      4136     1567 |      4.36   4.90       4.90 |
1.22   1.02   2.15      14.55 |   0.26   3.311
bench supply, no limit        5.25  0.10 |  18.11      8      4109     1690 |      4.21   5.22       5.21 |
1.17   0.96   2.10      14.50 |   0.24   3.311
bench supply, no limit        5.25  0.30 |  11.14      6      4104     1663 |      4.58   5.16       5.16 |
1.22   1.01   2.14      14.54 |   0.24   3.311
PC port, 1.0 A limit          4.75  0.10 |  16.20      7      4073     1395 |      1.61   4.70       4.68 |
1.72   1.51   2.65      15.05 |   0.27   3.311
PC port, 1.0 A limit          4.75  0.30 |  10.05      6      4077     1366 |      1.88   4.65       4.60 |
1.74   1.53   2.66      15.07 |   0.27   3.311
PC port, 1.0 A limit          5.00  0.10 |  17.05      8      4057     1515 |      1.68   4.95       4.90 |
1.69   1.48   2.62      15.02 |   0.25   3.311
PC port, 1.0 A limit          5.00  0.30 |  10.57      6      4057     1489 |      1.95   4.90       4.81 |
1.71   1.50   2.64      15.04 |   0.26   3.311
PC port, 1.0 A limit          5.25  0.10 |  17.90      7      4054     1635 |      1.75   5.21       5.01 |
1.66   1.45   2.59      14.99 |   0.24   3.311
PC port, 1.0 A limit          5.25  0.30 |  11.10      6      4052     1610 |      2.02   5.16       4.96 |
1.68   1.47   2.61      15.01 |   0.24   3.311
strict port, 0.55 A limit     4.75  0.10 |  16.16      7      4139     1460 |      1.45   4.70       3.21 |
2.62   2.00   3.13      15.58 |   0.27   3.311
strict port, 0.55 A limit     4.75  0.30 |  10.03      6      4132     1422 |      1.62   4.65       3.19 |
```

```
2.65   2.02   3.15      15.60 |   0.27   3.311
strict port, 0.55 A limit     5.00  0.10 |  17.01      7      4177     1635 |      1.52   4.95       3.21 |
2.57   1.94   3.08      15.52 |   0.25   3.311
strict port, 0.55 A limit     5.00  0.30 |  10.55      6      4168     1599 |      1.68   4.90       3.20 |
2.60   1.96   3.09      15.54 |   0.26   3.311
strict port, 0.55 A limit     5.25  0.10 |  17.85      7      4223     1804 |      1.59   5.21       3.22 |
2.51   1.89   3.02      15.47 |   0.24   3.311
strict port, 0.55 A limit     5.25  0.30 |  11.08      6      4212     1771 |      1.75   5.16       3.20 |
2.54   1.91   3.04      15.49 |   0.24   3.311

Q10ms: charge into J5 in the first 10 ms; Qexcess: the same less the steady current x 10 ms (the inrush part).
USB 2.0 7.2.4.1 attach limit for comparison: 10 uF and 50 uC. Board: 330 uF + ~14 uF ceramics on 5V -> about
1.7 mC to 5 V.

Wi-Fi TX burst on a settled board (C6 80 mA RX -> 354 mA TX), 5V at J5 and 3V3 dip:
  bench supply, no limit       Vs 4.75 Rcab 0.1: RX 4.60 V at 0.82 A, TX 4.56 V at 1.05 A; 3V3 dip about 11 mV
  bench supply, no limit       Vs 4.75 Rcab 0.3: RX 4.43 V at 0.85 A, TX 4.33 V at 1.10 A; 3V3 dip about 11 mV
  bench supply, no limit       Vs 5.00 Rcab 0.1: RX 4.86 V at 0.78 A, TX 4.82 V at 0.99 A; 3V3 dip about 11 mV
  bench supply, no limit       Vs 5.00 Rcab 0.3: RX 4.69 V at 0.80 A, TX 4.61 V at 1.04 A; 3V3 dip about 11 mV
  PC port, 1.0 A limit         Vs 4.75 Rcab 0.1: RX 4.60 V at 0.82 A, TX needs more than the port limit: 5V co
llapses; 3V3 dip about 11 mV
  PC port, 1.0 A limit         Vs 4.75 Rcab 0.3: RX 4.43 V at 0.85 A, TX needs more than the port limit: 5V co
llapses; 3V3 dip about 11 mV
  PC port, 1.0 A limit         Vs 5.00 Rcab 0.1: RX 4.86 V at 0.78 A, TX 4.82 V at 0.99 A; 3V3 dip about 11 mV
  PC port, 1.0 A limit         Vs 5.00 Rcab 0.3: RX 4.69 V at 0.80 A, TX needs more than the port limit: 5V co
llapses; 3V3 dip about 11 mV
  strict port, 0.55 A limit    Vs 4.75 Rcab 0.1: RX needs more than the port limit: 5V collapses, TX needs mor
e than the port limit: 5V collapses; 3V3 dip about 11 mV
  strict port, 0.55 A limit    Vs 4.75 Rcab 0.3: RX needs more than the port limit: 5V collapses, TX needs mor
e than the port limit: 5V collapses; 3V3 dip about 11 mV
  strict port, 0.55 A limit    Vs 5.00 Rcab 0.1: RX needs more than the port limit: 5V collapses, TX needs mor
e than the port limit: 5V collapses; 3V3 dip about 11 mV
  strict port, 0.55 A limit    Vs 5.00 Rcab 0.3: RX needs more than the port limit: 5V collapses, TX needs mor
e than the port limit: 5V collapses; 3V3 dip about 11 mV

Options, Vs 5.0 V, cable 0.3 ohm / 1 uH, host port bulk 120 uF (USB 2.0 minimum):
option                                              port                       load  |  Ipk A Vhost min
charge to 5V uC | 3V3 ms 1V2 ms 3V3 end
0 as built: R68 0.15 + C69 330 uF                   PC port, 1.0 A limit       BOOT  |  10.57      1.95
1720 |   1.71   2.64   3.311
0 as built: R68 0.15 + C69 330 uF                   PC port, 1.0 A limit       FULL  |  10.57      1.95
1720 |   1.71   2.64   2.335
0 as built: R68 0.15 + C69 330 uF                   strict port, 0.55 A limit  BOOT  |  10.55      1.68
1720 |   2.60   3.09   3.311
0 as built: R68 0.15 + C69 330 uF                   strict port, 0.55 A limit  FULL  |  10.55      1.68
1720 |  never  never   2.202
A soft-start load switch after J5, 5 ms ramp (C69 kept)    PC port, 1.0 A limit       BOOT  |   0.82      4.93
1720 |   3.73   4.66   3.311
A soft-start load switch after J5, 5 ms ramp (C69 kept)    PC port, 1.0 A limit       FULL  |   1.19      2.92
1720 |  never   4.66   2.335
A soft-start load switch after J5, 5 ms ramp (C69 kept)    strict port, 0.55 A limit  BOOT  |   0.65      4.44
1720 |   3.83   4.66   3.311
A soft-start load switch after J5, 5 ms ramp (C69 kept)    strict port, 0.55 A limit  FULL  |   0.85      2.47
1720 |  never  never   2.403
A2 same switch, 2 ms ramp                           PC port, 1.0 A limit       BOOT  |   1.02      4.90
1720 |   2.23   3.16   3.311
A2 same switch, 2 ms ramp                           PC port, 1.0 A limit       FULL  |   1.13      2.92
1720 |   2.23   3.16   2.335
A2 same switch, 2 ms ramp                           strict port, 0.55 A limit  BOOT  |   0.74      3.63
1720 |   2.48   3.41   3.311
```

```
A2 same switch, 2 ms ramp                                  strict port, 0.55 A limit  FULL  |   0.74      2.47
1720 |  never  never   1.236
B C69 47 uF polymer, R68 0.33 ohm                          PC port, 1.0 A limit       BOOT  |   9.94      3.77
305 |   1.17   2.10   3.311
B C69 47 uF polymer, R68 0.33 ohm                          PC port, 1.0 A limit       FULL  |   9.94      2.92
305 |   1.17   2.10   2.335
B C69 47 uF polymer, R68 0.33 ohm                          strict port, 0.55 A limit  BOOT  |   9.93      3.60
305 |   1.17   2.10   3.311
B C69 47 uF polymer, R68 0.33 ohm                          strict port, 0.55 A limit  FULL  |   9.93      2.49
305 |   1.17  never   2.587
C no damping branch (C69/R68 off), TVS for the ring        PC port, 1.0 A limit       BOOT  |   9.40      4.49
70 |   1.17   2.09   3.311
C no damping branch (C69/R68 off), TVS for the ring        PC port, 1.0 A limit       FULL  |   9.40      2.92
70 |   1.17   2.09   2.335
C no damping branch (C69/R68 off), TVS for the ring        strict port, 0.55 A limit  BOOT  |   9.40      4.46
70 |   1.17   2.09   3.311
C no damping branch (C69/R68 off), TVS for the ring        strict port, 0.55 A limit  FULL  |   9.40      2.49
70 |   1.17  never   2.283

r7f AS BUILT (parts read from design.json: {'u': 'U16', 'r1': 'R71', 'r2': 'R72', 'ilm': 'R73', 'cdv': 'C71',
'cin': ['C70']}): eFuse after J5 (UVLO 3.75 V on / 3.44 V off, ILIM 1.89 A (low end), dV/dt 0.42 V/ms, turn-on
delay 289 us), input capacitance 2.20 uF on VBUS_IN (1.21 uF at 5 V bias, assumed)
Attach transient (the first 339 us, before the eFuse conducts) against the USB 2.0 7.2.4.1 reference load 10 u
F || 44 ohm on the same port and cable:
port                           Vs  Rcab | Iatt A Qatt uC Vhost min | ref Iatt ref Q uC ref Vhost | VBUS_IN ma
x EN max
bench supply, no limit       4.75  0.10 |   4.78     6.5      4.68 |    11.65     83.6      4.36 |        8.6
7   2.77 OK
bench supply, no limit       4.75  0.30 |   4.11     7.3      4.69 |     8.25     82.9      4.47 |        7.4
9   2.40 OK
bench supply, no limit       5.25  0.10 |   5.29     7.1      5.17 |    12.88     92.4      4.81 |        9.5
8   3.06 OK
bench supply, no limit       5.25  0.30 |   4.55     7.9      5.18 |     9.12     91.7      4.94 |        8.2
8   2.65 OK
PC port, 1.0 A limit         4.75  0.10 |   4.78     6.5      4.68 |    11.64     83.3      4.22 |        8.6
7   2.77 OK
PC port, 1.0 A limit         4.75  0.30 |   4.11     7.3      4.69 |     8.24     82.8      4.38 |        7.4
9   2.40 OK
PC port, 1.0 A limit         5.25  0.10 |   5.29     7.1      5.17 |    12.87     92.1      4.66 |        9.5
8   3.06 OK
PC port, 1.0 A limit         5.25  0.30 |   4.55     7.9      5.18 |     9.12     91.5      4.83 |        8.2
8   2.65 OK
strict port, 0.55 A limit    4.75  0.10 |   4.78     6.5      4.67 |    11.62     83.2      4.19 |        8.6
7   2.77 OK
strict port, 0.55 A limit    4.75  0.30 |   4.11     7.3      4.68 |     8.23     82.6      4.35 |        7.5
0   2.40 OK
strict port, 0.55 A limit    5.25  0.10 |   5.29     7.1      5.16 |    12.85     91.8      4.63 |        9.5
8   3.06 OK
strict port, 0.55 A limit    5.25  0.30 |   4.55     7.9      5.18 |     9.11     91.3      4.80 |        8.2
8   2.65 OK
USB-C 1.5 A source           4.75  0.10 |   4.78     6.5      4.68 |    11.64     83.5      4.25 |        8.6
7   2.77 OK
USB-C 1.5 A source           4.75  0.30 |   4.11     7.3      4.69 |     8.25     82.9      4.41 |        7.4
9   2.40 OK
USB-C 1.5 A source           5.25  0.10 |   5.29     7.1      5.17 |    12.87     92.2      4.69 |        9.5
8   3.06 OK
USB-C 1.5 A source           5.25  0.30 |   4.55     7.9      5.18 |     9.12     91.6      4.86 |        8.2
8   2.65 OK
USB-C 3.0 A source           4.75  0.10 |   4.78     6.5      4.68 |    11.66     83.6      4.31 |        8.6
7   2.77 OK
```

```
USB-C 3.0 A source           4.75  0.30 |   4.11     7.3      4.69 |     8.25     82.9      4.46 |        7.4
9   2.40 OK
USB-C 3.0 A source           5.25  0.10 |   5.29     7.1      5.17 |    12.89     92.3      4.76 |        9.5
8   3.06 OK
USB-C 3.0 A source           5.25  0.30 |   4.55     7.9      5.18 |     9.12     91.7      4.93 |        8.2
8   2.65 OK
attach transient no worse than the 10 uF / 50 uC reference in every corner: YES; VBUS_IN must stay under 20 V
(IN abs max), EN under 7 V (abs max)

Cold start through the eFuse (0-60 ms):
port                         load     Vs  Rcab |  Ipk A Vhost min 5V max 5V end | 3V3 ms 1V2 ms  PU ms 3V3 end
| U2 starts EN starts in ILIM ms FET Wpk FET mJ
bench supply, no limit       BOOT   4.75  0.10 |   0.61      4.68   4.69   4.69 |   8.56   8.37  20.94   3.311
|         1         4       0.00    1.09    5.7 starts
bench supply, no limit       BOOT   4.75  0.30 |   0.61      4.69   4.63   4.63 |   8.54   8.36  20.93   3.311
|         1         2       0.00    1.01    5.4 starts
bench supply, no limit       BOOT   5.25  0.10 |   0.61      5.17   5.19   5.19 |   8.56   8.37  20.94   3.311
|         1         4       0.00    1.39    7.4 starts
bench supply, no limit       BOOT   5.25  0.30 |   0.61      5.18   5.15   5.15 |   8.54   8.35  20.93   3.311
|         1         2       0.00    1.32    7.0 starts
USB-C 3.0 A source           BOOT   4.75  0.10 |   0.61      4.68   4.69   4.69 |   8.56   8.37  20.94   3.311
|         1         4       0.00    1.09    5.7 starts
USB-C 3.0 A source           BOOT   4.75  0.30 |   0.61      4.69   4.63   4.63 |   8.54   8.36  20.93   3.311
|         1         2       0.00    1.01    5.4 starts
USB-C 3.0 A source           BOOT   5.25  0.10 |   0.61      5.17   5.19   5.19 |   8.56   8.37  20.94   3.311
|         1         4       0.00    1.39    7.4 starts
USB-C 3.0 A source           BOOT   5.25  0.30 |   0.61      5.18   5.15   5.15 |   8.54   8.35  20.93   3.311
|         1         2       0.00    1.32    7.0 starts
USB-C 1.5 A source           BOOT   4.75  0.10 |   0.61      4.68   4.69   4.69 |   8.56   8.37  20.94   3.311
|         1         4       0.00    1.09    5.7 starts
USB-C 1.5 A source           BOOT   4.75  0.30 |   0.61      4.69   4.63   4.63 |   8.54   8.36  20.93   3.311
|         1         2       0.00    1.01    5.4 starts
USB-C 1.5 A source           BOOT   5.25  0.10 |   0.61      5.17   5.19   5.19 |   8.56   8.37  20.94   3.311
|         1         4       0.00    1.39    7.4 starts
USB-C 1.5 A source           BOOT   5.25  0.30 |   0.61      5.18   5.15   5.15 |   8.54   8.35  20.93   3.311
|         1         2       0.00    1.32    7.0 starts
PC port, 1.0 A limit         BOOT   4.75  0.10 |   0.61      4.68   4.69   4.69 |   8.56   8.37  20.94   3.311
|         1         4       0.00    1.09    5.7 starts
PC port, 1.0 A limit         BOOT   4.75  0.30 |   0.61      4.69   4.63   4.63 |   8.54   8.36  20.93   3.311
|         1         2       0.00    1.01    5.4 starts
PC port, 1.0 A limit         BOOT   5.25  0.10 |   0.61      5.17   5.19   5.19 |   8.56   8.37  20.94   3.311
|         1         4       0.00    1.39    7.4 starts
PC port, 1.0 A limit         BOOT   5.25  0.30 |   0.61      5.18   5.15   5.15 |   8.54   8.35  20.93   3.311
|         1         2       0.00    1.32    7.0 starts
strict port, 0.55 A limit    BOOT   4.75  0.10 |   0.61      4.67   4.69   4.69 |   8.56   8.37  20.94   3.311
|         1         4       0.00    1.07    5.7 starts
strict port, 0.55 A limit    BOOT   4.75  0.30 |   0.61      4.68   4.63   4.63 |   8.54   8.35  20.93   3.311
|         1         2       0.00    1.00    5.4 starts
strict port, 0.55 A limit    BOOT   5.25  0.10 |   0.61      5.16   5.19   5.19 |   8.56   8.37  20.94   3.311
|         1         4       0.00    1.38    7.4 starts
strict port, 0.55 A limit    BOOT   5.25  0.30 |   0.61      5.18   5.15   5.15 |   8.54   8.35  20.93   3.311
|         1         2       0.00    1.31    7.0 starts
bench supply, no limit       FULL   4.75  0.10 |   1.51      4.63   4.51   4.51 |   8.66   8.37  20.96   3.311
|         1         4       0.00    2.06   11.1 starts
bench supply, no limit       FULL   4.75  0.30 |   1.51      4.63   4.27   4.27 |   8.65   8.35  20.95   3.311
|         1         2       0.00    1.77    9.9 starts
bench supply, no limit       FULL   5.25  0.10 |   1.51      5.13   5.03   5.03 |   8.66   8.37  20.96   3.311
|         1         4       0.00    2.67   14.2 starts
bench supply, no limit       FULL   5.25  0.30 |   1.51      5.13   4.83   4.83 |   8.65   8.35  20.95   3.311
|         1         2       0.00    2.38   12.5 starts
```

```
USB-C 3.0 A source          FULL   4.75  0.10 |   1.51      4.63   4.51   4.51 |   8.66   8.37  20.96   3.311
|         1         4       0.00    2.06   11.1 starts
USB-C 3.0 A source          FULL   4.75  0.30 |   1.51      4.63   4.27   4.27 |   8.65   8.35  20.95   3.311
|         1         2       0.00    1.77    9.9 starts
USB-C 3.0 A source          FULL   5.25  0.10 |   1.51      5.13   5.03   5.03 |   8.66   8.37  20.96   3.311
|         1         4       0.00    2.67   14.2 starts
USB-C 3.0 A source          FULL   5.25  0.30 |   1.51      5.13   4.83   4.83 |   8.65   8.35  20.95   3.311
|         1         2       0.00    2.38   12.5 starts
USB-C 1.5 A source          FULL   4.75  0.10 |   1.51      4.63   4.51   4.51 |   8.66   8.37  20.96   3.311
|         1         4       0.00    2.06   11.1 starts
USB-C 1.5 A source          FULL   4.75  0.30 |   1.51      4.63   4.27   4.27 |   8.65   8.35  20.95   3.311
|         1         2       0.00    1.77    9.9 starts
USB-C 1.5 A source          FULL   5.25  0.10 |   1.51      5.13   5.03   5.03 |   8.66   8.37  20.97   3.311
|         1         4       0.00    2.67   14.2 starts
USB-C 1.5 A source          FULL   5.25  0.30 |   1.51      5.13   4.83   4.83 |   8.65   8.35  20.95   3.311
|         1         2       0.00    2.38   12.5 starts
PC port, 1.0 A limit        FULL   4.75  0.10 |   1.25      3.57   3.14   2.40 |  never  never  never   0.000
|         7        49       0.00    1.96   16.1 DOES NOT START CLEANLY
PC port, 1.0 A limit        FULL   4.75  0.30 |   1.22      3.80   3.09   2.40 |  never  never  never   0.000
|         7        38       0.00    1.67   13.8 DOES NOT START CLEANLY
PC port, 1.0 A limit        FULL   5.25  0.10 |   1.31      3.57   3.23   2.40 |  never  never  never   0.000
|         7        54       0.00    2.57   22.3 DOES NOT START CLEANLY
PC port, 1.0 A limit        FULL   5.25  0.30 |   1.28      3.82   3.18   2.40 |  never  never  never   0.000
|         7        38       0.00    2.27   19.6 DOES NOT START CLEANLY
strict port, 0.55 A limit   FULL   4.75  0.10 |   1.21      3.55   2.85   2.40 |  never  never  never   0.001
|         8        44       0.00    1.36   10.3 DOES NOT START CLEANLY
strict port, 0.55 A limit   FULL   4.75  0.30 |   1.17      3.79   2.83   2.40 |  never  never  never   0.000
|         8        27       0.00    1.24    9.3 DOES NOT START CLEANLY
strict port, 0.55 A limit   FULL   5.25  0.10 |   1.21      3.55   2.90   2.40 |  never  never  never   0.027
|         8        51       0.00    1.81   13.7 DOES NOT START CLEANLY
strict port, 0.55 A limit   FULL   5.25  0.30 |   1.21      3.77   2.88   2.40 |  never  never  never   0.003
|         8        37       0.00    1.65   12.5 DOES NOT START CLEANLY
full load starts cleanly (one U2 start, 3V3 at 95 %, CHIP_PU released) on every >= 1.5 A source corner: YES
full-load peak during the ramp 1.51 A; verify.py gates ILIM low end > 1.51 A x 1.1: consistent
Hot-plug overshoot at U2 VIN (tools/hotplug.py grid, ceramics only, no TVS): 0 as built 5.34 V, B 47 uF / 0.33
6.99 V, C no damping 9.83 V; limit 5.8 V (SY8089 abs max 6 V)
```

附錄 11：impedance_r7f.txt

```
stackup read from the board:
  F.SilkS        Top Silk Screen
  F.Paste        Top Solder Paste
  F.Mask         Top Solder Mask        0.01
  F.Cu           copper                 0.035
  dielectric 1   prepreg                0.0994   3313   4.1
  In1.Cu         copper                 0.0152
  dielectric 2   core                   0.35     FR4    4.6
  In2.Cu         copper                 0.0152
  dielectric 3   prepreg                0.1088   2116   4.16
  In3.Cu         copper                 0.0152
  dielectric 4   core                   0.35     FR4    4.6
  In4.Cu         copper                 0.0152
  dielectric 5   prepreg                0.0994   3313   4.1
  B.Cu           copper                 0.035
  B.Mask         Bottom Solder Mask     0.01
  B.Paste        Bottom Solder Paste
  B.SilkS        Bottom Silk Screen
solver uses h 0.0994 er 4.1 t 0.035 for F/In1 and B/In4: same as the board stackup
copper + dielectric 1.1384 mm (order: 1.1384)

Method 1 (field solver) and method 2 (closed form, no mask; Kirschning-Jansen is for zero-thickness copper, se
e the check below). Ohm.
line                                    solver mask  solver no mask      K-J t=0  IPC-2141A  target
TMDS, CSI: 0.127 / 0.198, F ref In1, B ref In4         99.1           105.4        121.2      101.3  100 +-10%
OK  (5.72 ps/mm)
TMDS ESD flow-through 0.127 / 0.373             105.9           112.0        126.5      107.6  100 +-10%  OK
(5.81 ps/mm)
USB HS: 0.18 / 0.32                              89.1            93.3        103.5       87.0  90 +-10%  OK
(5.84 ps/mm)
LNA_OUT 0.15, pour gap 0.25 (CPWG)               50.5            53.0         52.9  (no pour)  50 +-10%  OK

The closed forms are for a zero-thickness strip; at t/h = 0.35 that is not this board. So the coupled formula
checks the solver with the
copper made thin (t 0.5 um), where both describe the same geometry:
  TMDS, CSI: 0.127 / 0.198, F ref In1, B ref In4 solver thin  118.8   Kirschning-Jansen  121.2   difference -
2.0%
  TMDS ESD flow-through 0.127 / 0.373      solver thin  123.9   Kirschning-Jansen  126.5   difference -2.0%
  USB HS: 0.18 / 0.32                      solver thin  101.6   Kirschning-Jansen  103.5   difference -1.8%

Sensitivity of TMDS/CSI 0.127 / 0.198 (solver, mask), pitch held at 0.325:
  w 0.107: 107.6
  w 0.147: 91.5
  h 0.0894: 94.7
  h 0.1094: 102.8
  er 3.8: 101.8
  er 4.4: 96.5

In2 (In3 3V3 below, In1 GND above), solver; In2 has no impedance-controlled line in r7. H_MID 0.1164 in the so
lver against 0.1088 ordered:
  w 0.127 single-ended, prepreg 0.1088: 49.5 ohm
  w 0.127 single-ended, prepreg 0.1164: 50.9 ohm
IMPEDANCE ALL ON TARGET
```

附錄 12：s3_solver_test_r7f.txt

```
PASS Cohn stripline w=0.1234567 b=0.4: closed form 127.75, solver 126.09 (-1.30 %)
PASS Cohn stripline w=0.2003333 b=0.5: closed form 112.83, solver 111.72 (-0.99 %)
PASS Cohn coupled stripline w=0.1333333 s=0.1987654 b=0.5 (Zdiff): closed form 238.87, solver 235.59 (-1.37 %)
PASS H-J microstrip w=0.1777777 h=0.1 er=4.1: closed form 53.95, solver 53.40 (-1.02 %)
PASS continuity w 0.12267345678 g 0.2025265432 h 0.0994 against 0.1227/0.2025/0.0994: Zdiff 100.83 / 100.82 oh
m (+0.01 %, limit 0.5 %)
PASS continuity w 0.127 g 0.198 h 0.10123456789 against 0.127/0.198/0.1012346: Zdiff 99.80 / 99.80 ohm (+0.00
%, limit 0.5 %)
RESULT PASS
```

附錄 13：pairs_r7f.txt

```
out/p4_kvm_mini.kicad_pcb
pair          P mm     N mm  skew mm     ps    %UI vias P/N  90deg  acute  widths
D2           8.936    8.933    0.003    0.0   0.00     0/0       0      0  [0.127]
D1          10.750   11.083    0.333    1.9   0.31     0/0       0      0  [0.127]
D0          13.369   13.373    0.003    0.0   0.00     0/0       0      0  [0.127]
CK          15.426   15.523    0.096    0.6   0.09     0/0       0      0  [0.127]
CSI_D0      10.103   10.103    0.000    0.0   0.00     1/1       0      0  [0.127]
CSI_D1      10.062   10.148    0.085    0.5   0.05     1/1       0      0  [0.127]
CSI_CK      10.156   10.247    0.091    0.5   0.05     1/1       0      0  [0.127]
```

附錄 14：pairgap_r7f.md

```
# Differential pair gap and coupling (tools/pairgap.py)

Differential-pair gap and coupling (r7e, REVIEW_r7d: verify.py never checked pair spacing).

Each line of a pair is sampled every 0.05 mm; at each sample the edge-to-edge gap to the partner line ON THE S
AME LAYER is measured and
turned into a differential impedance with the field solver's table below (tools/r7/impedance_r7.py's solver, t
his stackup, solder mask,
2026-11-09). This board's pairs are weakly coupled (outer dielectric 0.0994 mm), so an uncoupled stretch is on
ly ~10% above target while a
tight one drops fast.
  FAIL  any sample whose impedance is outside target +-10% (CSI/TMDS 100 ohm: gap under 0.12 mm; USB HS 90 oh
m: gap under 0.14 mm),
        or a line with no partner segment on its layer for more than half its length.
  report  coupled length (gap within +-0.03 mm of the solved gap) and uncoupled length (fan-outs at balls, pad
s, vias, matching bumps).

| group | line | partner | solved gap | length mm | coupled mm | uncoupled mm | no partner on layer mm | Zdiff
min-max ohm | out of +-10% |
|---|---|---|---|---|---|---|---|---|---|
| csi | CSI_D0_P | CSI_D0_N | 0.198 | 10.10 | 4.54 | 5.56 | 0.00 | 99.0-109.5 | 0  |
| csi | CSI_D0_N | CSI_D0_P | 0.198 | 10.10 | 4.90 | 5.20 | 0.00 | 99.0-109.0 | 0  |
| csi | CSI_D1_P | CSI_D1_N | 0.198 | 10.06 | 7.51 | 2.55 | 0.00 | 99.0-108.4 | 0  |
| csi | CSI_D1_N | CSI_D1_P | 0.198 | 10.15 | 7.73 | 2.41 | 0.00 | 99.0-108.0 | 0  |
| csi | CSI_CK_P | CSI_CK_N | 0.198 | 10.16 | 7.42 | 2.73 | 0.00 | 99.0-109.5 | 0  |
| csi | CSI_CK_N | CSI_CK_P | 0.198 | 10.25 | 6.94 | 3.31 | 0.00 | 99.0-109.2 | 0  |
| tmds | D0_P | D0_N | 0.198 | 13.37 | 7.59 | 5.78 | 0.00 | 99.0-109.5 | 0  |
| tmds | D0_N | D0_P | 0.198 | 13.37 | 8.05 | 5.32 | 0.00 | 99.0-109.0 | 0  |
| tmds | D1_P | D1_N | 0.198 | 10.75 | 5.59 | 5.16 | 0.00 | 99.0-108.7 | 0  |
| tmds | D1_N | D1_P | 0.198 | 11.08 | 6.05 | 5.03 | 0.00 | 99.0-109.0 | 0  |
| tmds | D2_P | D2_N | 0.198 | 8.94 | 3.75 | 5.19 | 0.00 | 99.0-108.7 | 0  |
| tmds | D2_N | D2_P | 0.198 | 8.93 | 4.40 | 4.53 | 0.00 | 99.0-109.0 | 0  |
| tmds | CK_P | CK_N | 0.198 | 15.43 | 9.09 | 6.34 | 0.00 | 99.0-109.5 | 0  |
| tmds | CK_N | CK_P | 0.198 | 15.52 | 9.55 | 5.97 | 0.00 | 99.0-109.0 | 0  |
| usb | USB_DP | USB_DN | 0.32 | 9.67 | 5.55 | 4.12 | 0.00 | 86.8-93.0 | 0  |
| usb | USB_DN | USB_DP | 0.32 | 9.50 | 5.85 | 3.65 | 0.00 | 86.8-92.4 | 0  |
RESULT PASS
```

附錄 15：refplane_r7f.txt

```
warn CSI_CK_N  F.Cu gap 0.0017 mm2 within 0.1 mm at (5.167, -3.132); other-net vias within 0.8 mm: [(0.31, 'CS
I_D1_N')]
warn CSI_CK_N  B.Cu gap 0.0009 mm2 within 0.1 mm at (4.745, -3.119); other-net vias within 0.8 mm: [(0.31, 'CS
I_D1_N'), (0.5, 'GND'), (0.61, 'GND')]
warn CSI_CK_P  B.Cu gap 0.0021 mm2 within 0.1 mm at (5.200, -3.167); other-net vias within 0.8 mm: [(0.31, 'CS
I_D1_N')]
warn D0_N      B.Cu gap 0.0128 mm2 within 0.1 mm at (-7.388, -3.983); other-net vias within 0.8 mm: [(0.29, 'V
DDC')]
warn D1_N      B.Cu gap 0.0101 mm2 within 0.1 mm at (-5.393, -4.636); other-net vias within 0.8 mm: [(0.29, 'D
DC_SCL')]
warn CK_N      B.Cu gap 0.0021 mm2 within 0.1 mm at (1.450, -2.883); other-net vias within 0.8 mm: [(0.31, 'VD
DC')]
warn CK_P      B.Cu gap 0.0060 mm2 within 0.1 mm at (-7.689, -4.261); other-net vias within 0.8 mm: [(0.3, 'VD
DC')]
warn USB_DN    B.Cu gap 0.0187 mm2 within 0.1 mm at (4.662, -9.215); other-net vias within 0.8 mm: [(0.28, 'US
B2_DN'), (0.73, 'GND'), (0.79, 'GND')]
warn USB_DN    B.Cu gap 0.0062 mm2 within 0.1 mm at (4.729, -9.429); other-net vias within 0.8 mm: [(0.3, 'USB
2_DN')]
csi tmds usb: 0 gaps within 0.03 mm of the lines (FAIL), 9 within 0.1 mm (listed), 0.0607 mm2
RESULT PASS
```

附錄 16：mask_report.txt

```
mask bridge minimum 0.1088 mm, 0 under 0.1, 3 under 0.13 (black/white only); opening to other-net conductor mi
nimum 0.1007 mm, 0 under 0.09; 0 openings merged across nets; openings {'F': {'openings': 213, 'copper': 977,
'not_on_a_pad': 0}, 'B': {'openings': 360, 'copper': 1491, 'not_on_a_pad': 0}}
bridges under 0.10: 0
bridges 0.10 to 0.13: 3
  B 0.109 J5.1 [GND] | C64.2 [GND] near (12.52,6.82)
  B 0.111 J5.3 [GND] | R72.1 [GND] near (16.70,6.82)
  B 0.122 J7.SH1 [GND] | R78.2 [SD_D2_C] near (-19.12,-1.47)
openings merged across nets: 0
opening to conductor under 0.09: 0
RESULT PASS
```

附錄 17：hole_report.txt

```
0 under limit; VIA outer min 0.2 (limit 0.2), VIA inner min 0.2012 (limit 0.2), NPTH outer min none near (limi
t 0.2), NPTH inner min none near (limit 0.2), PTH outer min 0.3017 (limit 0.28), PTH inner min 0.32 (limit 0.
3); holes {'VIA': 318, 'PTH': 16, 'NPTH': 0} on 6 copper layers
limits: JLCPCB capabilities page, 2026-11-06. Same-net copper on plated holes is exempt, NPTH has no exemptio
n.
same-net objects exempted, by hole type and object: {('PTH', 'ComponentPad'): 96, ('PTH', 'Conductor'): 8, ('P
TH', 'SMDPad'): 2, ('PTH', 'ViaPad'): 24, ('PTH', 'Zone'): 65, ('VIA', 'ComponentPad'): 30, ('VIA', 'Conducto
r'): 728, ('VIA', 'SMDPad'): 213, ('VIA', 'ViaPad'): 2124, ('VIA', 'Zone'): 528}
zone regions whose parsed area was checked against the outline: 27
PTH holes, closest other-net copper within 0.45 mm of the hole edge: 16 objects (9 under the 0.35 JLC recommen
ds)
  0.3017 limit 0.28 J5.2 [GND] L6 to track(13.74,-4.73) [V1V2_SNS]
  0.3200 limit 0.3 J3.4 [GND] L4 to zone(17.36,-19.53) [3V3]
  0.3200 limit 0.3 J5.1 [GND] L4 to zone(17.36,-19.53) [3V3]
  0.3200 limit 0.3 J3.3 [GND] L4 to zone(17.36,-19.53) [3V3]
  0.3200 limit 0.3 J3.1 [GND] L4 to zone(17.36,-19.53) [3V3]
  0.3200 limit 0.3 J5.4 [GND] L4 to zone(17.36,-19.53) [3V3]
  0.3200 limit 0.3 J5.3 [GND] L4 to zone(17.36,-19.53) [3V3]
  0.3205 limit 0.3 J3.2 [GND] L4 to zone(17.36,-19.53) [3V3]
  0.3205 limit 0.3 J5.2 [GND] L4 to zone(17.36,-19.53) [3V3]
  0.3534 limit 0.3 J5.4 [GND] L3 to track(15.59,11.19) [HOST_VBUS_DIV]
  0.3534 limit 0.3 J5.3 [GND] L3 to track(15.59,11.19) [HOST_VBUS_DIV]
  0.3705 limit 0.3 J2.22 [GND] L4 to zone(17.36,-19.53) [3V3]
NPTH hole wall to any copper, reported margin above the 0.2 limit. 0 holes:
  no copper within 0.45 mm
under limit: 0
RESULT PASS
```

附錄 18：bodycheck_r7f.txt（body_report.txt 內容相同）

```
Real outline check. mm. See tools/bodycheck.py for what each gap means.
ASSUMPTION: worst = max less +-0.1 mm placement on each part, in x and y. JLC publishes no number.
flash = worst, with TI DBV0005A note 4 at its 0.25 mm per side limit. Reported, not used to refuse.
Height is not considered.

RESULT PASS 129 parts, 4908 same-side pairs, 0 overlaps, 0 worst-case warnings, smallest nominal gap 0.350 mm

Overlaps (refused):
  none

Worst-case warnings (nominal and max both clear):
  none

Closest 40 pairs:
  C19/C20      B  nom  0.400 (area  0.00)  max  0.250  worst  0.050  flash       -  body-to-land  0.325
  C6/R79       F  nom  0.350 (area  0.00)  max  0.250  worst  0.050  flash       -  body-to-land  0.290
  U7/C17       B  nom  0.400 (area  0.00)  max  0.250  worst  0.050  flash       -  body-to-land  0.140
  C61/C62      B  nom  0.650 (area  0.00)  max  0.250  worst  0.050  flash       -  body-to-land  0.600
  U7/C24       B  nom  0.410 (area  0.00)  max  0.260  worst  0.060  flash       -  body-to-land  0.350
  R76/R77      B  nom  0.360 (area  0.00)  max  0.260  worst  0.060  flash       -  body-to-land  0.300
  L1/C61       B  nom  0.675 (area  0.00)  max  0.275  worst  0.075  flash       -  body-to-land  0.625
  U6/R28       B  nom  0.429 (area  0.00)  max  0.284  worst  0.080  flash       -  body-to-land  0.320
  R60/U7       B  nom  0.430 (area  0.00)  max  0.280  worst  0.080  flash       -  body-to-land  0.370
  U2/L1        B  nom  0.555 (area  0.00)  max  0.280  worst  0.080  flash       -  body-to-land  0.180
  L1/U12       B  nom  0.690 (area  0.00)  max  0.290  worst  0.090  flash       -  body-to-land  0.840
  C10/Q1       B  nom  0.595 (area  0.00)  max  0.375  worst  0.094  flash       -  body-to-land  0.330
  R61/C68      F  nom  0.400 (area  0.00)  max  0.300  worst  0.100  flash       -  body-to-land  0.340
  R11/C5       F  nom  0.500 (area  0.00)  max  0.335  worst  0.100  flash       -  body-to-land  0.342
  R61/R63      F  nom  0.400 (area  0.00)  max  0.300  worst  0.100  flash       -  body-to-land  0.340
  C16/R25      B  nom  0.500 (area  0.00)  max  0.300  worst  0.100  flash       -  body-to-land  0.240
  C16/R26      B  nom  0.500 (area  0.00)  max  0.300  worst  0.100  flash       -  body-to-land  0.240
  C11/U10      B  nom  0.600 (area  0.00)  max  0.300  worst  0.100  flash       -  body-to-land  0.525
  U8/C20       B  nom  0.450 (area  0.00)  max  0.300  worst  0.100  flash       -  body-to-land  0.070
  Q1/R45       B  nom  0.471 (area  0.00)  max  0.300  worst  0.100  flash       -  body-to-land  0.204
  U7/C21       B  nom  0.450 (area  0.00)  max  0.300  worst  0.100  flash       -  body-to-land  0.190
  Q1/R44       B  nom  0.460 (area  0.00)  max  0.300  worst  0.100  flash       -  body-to-land  0.200
  Q1/R43       B  nom  0.549 (area  0.00)  max  0.361  worst  0.100  flash       -  body-to-land  0.312
  R72/R71      B  nom  0.400 (area  0.00)  max  0.300  worst  0.100  flash       -  body-to-land  0.340
  FB2/FB3      B  nom  0.710 (area  0.00)  max  0.310  worst  0.110  flash       -  body-to-land  0.660
  C16/R27      B  nom  0.673 (area  0.00)  max  0.424  worst  0.141  flash       -  body-to-land  0.431
  C10/U10      B  nom  0.849 (area  0.00)  max  0.424  worst  0.141  flash       -  body-to-land  0.569
  C64/U2       B  nom  0.467 (area  0.00)  max  0.350  worst  0.150  flash       -  body-to-land  0.350
  C66/C67      F  nom  0.750 (area  0.00)  max  0.350  worst  0.150  flash       -  body-to-land  0.700
  FB3/FB4      B  nom  0.750 (area  0.00)  max  0.350  worst  0.150  flash       -  body-to-land  0.700
  U7/C15       B  nom  0.500 (area  0.00)  max  0.350  worst  0.150  flash       -  body-to-land  0.240
  C21/R21      B  nom  0.450 (area  0.00)  max  0.350  worst  0.150  flash       -  body-to-land  0.390
  C13/C16      B  nom  0.500 (area  0.00)  max  0.350  worst  0.150  flash       -  body-to-land  0.240
  R80/R81      F  nom  0.400 (area  0.00)  max  0.370  worst  0.170  flash       -  body-to-land  0.280
  U7/C22       B  nom  0.550 (area  0.00)  max  0.400  worst  0.200  flash       -  body-to-land  0.490
  U7/C23       B  nom  0.550 (area  0.00)  max  0.400  worst  0.200  flash       -  body-to-land  0.490
  U7/C31       B  nom  0.550 (area  0.00)  max  0.400  worst  0.200  flash       -  body-to-land  0.490  not f
itted
  U7/C18       B  nom  0.600 (area  0.00)  max  0.400  worst  0.200  flash       -  body-to-land  0.220
  J7/R78       B  nom  0.545 (area  0.00)  max  0.420  worst  0.220  flash       -  body-to-land  0.367
  U16/C71      B  nom  0.550 (area  0.00)  max  0.450  worst  0.250  flash       -  body-to-land  0.290
```

```
Sources, one line per outline used:
  U1: Waveshare wiki ESP32-P4-Module, Dimensions drawing: 25.00 x 25.00, H 3.30 + 0.80. Tolerance not publishe
d, +-0.15 ASSUMED
  J8: Hirose U.FL catalogue p3: 3.0 x 2.6 body, 3.1 over the tabs, centred on the mating axis. Tolerance not l
isted, +-0.1 ASSUMED
  J3,J5: HX TYPE-C 16P L8.35: JLC lists no drawing. Outline from the footprint model, UNVERIFIED. +-0.1 ASSUME
D
  U3,U14,U8,U9: TI DQA0010A (TPD4E05U06 p28): 2.4-2.6 x 0.9-1.1, no leads
  R1,R2,R3,R4,R60,R9,R10,R11,R12,R13,R20,R21,R22,R23,R24,R25,R26,R27,R28,R30,R31,R32,R40,R41,R42,R43,R44,R45,R
46,R47,R50,R51,R52,R53,R54,R55,R61,R62,R63,R65,R66,R67,R72,R71,R73,R76,R77,R78,R74,R75,R79: UNI-ROYAL 0402 p4:
1.0+-0.10 x 0.5+-0.05
  C50,C3,C4,C5,C6,C13,C15,C17,C19,C21,C22,C23,C24,C25,C26,C27,C29,C30,C31,C32,C33,C41,C64,C63,C68,C70,C71: Sam
sung CL05 p6 / FH 0402B: 1.0+-0.05 x 0.5+-0.05
  J6: JST eSH p1,p3: B 10, depth 4.25, tails 0.7, height 2.9. Housing +-0.1, tail width 0.3 and tab size insid
e its land are ASSUMED
  C1,C2,C10,C11,C12,C14,C16,C18,C20,C40: Samsung CL10 p6: 1.6+-0.1 x 0.8+-0.1
  J2: XUNPU HDMI-001S drawing: width 14.95. Depth from the existing footprint model, not re-measured. +-0.1 AS
SUMED
  U7: TC358743XBG, P-VFBGA64-0606: 6.0 square. Tolerance +-0.1 ASSUMED
  U4: TI DBV0005A (TLV733 SBVS235 p36): 2.75-3.05 x 1.45-1.75, span 2.6-3.0, lead 0.3-0.5. Note 4: up to 0.25
flash per side not included
  FB2,FB4: Sunlord GZ series: 2.0+-0.2 x 1.25+-0.2 (EIA 0805, LCSC scan not read at table level)
  FB3,FB5: Murata BLM21 series (JENF243A-0005Y-01): 2.0+-0.2 x 1.25+-0.2, EIA 0805
  Y1: YXC YSO110TR p3: 3.2+-0.1 x 2.5+-0.1, H 1.2 max
  FB6: Murata BLM18AG JENF243A p4: 1.6+-0.15 x 0.8+-0.15
  U6: TI DGK0008A (TCA9517A SCPS245E p22): 2.9-3.1 square, span 4.75-5.05, lead 0.25-0.38. Notes 3, 4: up to
0.15 mold flash and 0.25 interlead flash not included
  J7: Molex SD-104031-001 rev B2 sheet 1: 11.95+-0.15 x 11.40+-0.15, H 1.42+-0.15
  Q1: NOT THE MAKER'S DRAWING: AOS AO3401A Rev 3.1 has no outline and none was found. Stand-in TI DBZ0003A (JE
DEC TO-236): 2.80-3.04 x 1.2-1.4, span 2.10-2.64, lead 0.3-0.5
  C60,C61,C62,C65,C66,C67: Samsung CL series p6, 0805: 2.00+-0.20 x 1.25+-0.20 at most
  U2,U15: Silergy SY8089 AN Rev 0.9A p9: 2.80-3.10 x 1.50-1.70, span 2.70-3.00, lead 0.30-0.50
  L1,L2: Sunlord SWPA series catalogue, SWPA4020S row: 4.0+-0.2 square, H 2.0 max
  J4: JST eSH p1,p3: B 8, depth 4.25, tails 0.7, height 2.9. Housing +-0.1, tail width 0.3 and tab size inside
its land are ASSUMED
  U10,U11,U12: Everlight EL3H7-G DPC-0000031 rev 7 p7: 4.40+-0.2 x 2.70+-0.2, span 7.00+-0.3, lead 0.40+-0.1,
pitch 1.27, H 2.0 max
  C69: Panasonic 10TPE330M: 7.3 x 4.3 per LCSC listing; Panasonic drawing not obtained, so EIA 7343 +-0.3 x +-
0.3 ASSUMED
  R68: UNI-ROYAL 0805W8F: EIA 0805 2.0+-0.15 x 1.25+-0.15 (UNI-ROYAL thick film series table, not re-read for
r7c)
  U16: TI DSG0008A (TPS2595 SLVSE57C p45): 2.0+-0.1 square, no leads (pads under the body)
  D1: TI DPY0002A (TPD1E10B06 SLLSEB1G p19): 1.0+-0.05 x 0.6+-0.05, H 0.45 max
  R80,R81: UNI-ROYAL RC-01W 0201 (LCSC listing): 0.60+-0.03 x 0.30+-0.03; maker drawing not read, EIA 0201 ASS
UMED
```

附錄 19：CPL_polarity_pin1_r7f.md

```
# CPL rotation and pin 1 of polarized parts (tools/cpl_pin1.py)

CPL rotation and pin 1 of every polarized part (r7e, REVIEW_r7d M1).

For each polarized part on the CPL (ICs, connectors, the crystal oscillator, polarized capacitors; plain R/C/
L/FB/TP are not polarized):
  - pin 1 (A1 for the BGA, A1B12 for USB-C) from design.json, as an offset from the part centre in the JLC top
preview (x right, y down,
    = KiCad screen) and in the JLC bottom preview (x mirrored);
  - pin 1 in the footprint's own frame (rotation and bottom mirror undone; the convention is checked: every pa
rt of one footprint gives the
    same vector whatever its angle and side);
  - the JLC angle offset (CPL = KiCad + offset) from data/jlc_offsets.json. Only parts a person verified in JL
C's preview have one; for every
    other part the offset is UNKNOWN (not 0) and the row says CHECK IN PREVIEW.
FAIL when: a verified part's CPL angle is not KiCad + offset; its pin 1 (footprint frame) is not where it was
when verified (footprint
changed); a polarized part is missing from the CPL; a polarized capacitor has pin 1 on GND or pin 2 off GND, o
r its + pin does not reach a
positive supply through at most one resistor (C69: + on 5V_DAMP, R68 to 5V, REVIEW_r7d M1).

| Ref | LCSC | Land | Side | KiCad deg | CPL deg | JLC offset | Pin 1 | Net | Pin 1 in JLC top view (mm) | Pin
1 in JLC bottom view | Status |
|---|---|---|---|---|---|---|---|---|---|---|---|
| C69 | C347552 | CP_7343 | Bottom | 0 | 0 | UNKNOWN | 1 | 5V_DAMP | right (+3.10, +0.00) | left (-3.10, +0.0
0) | CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| D1 | C48260 | X1SON-2_0.6x1 | Top | 0 | 0 | UNKNOWN | 1 | HDMI_CEC | left (-0.35, +0.00) | - | CHECK IN PREV
IEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| J2 | C720616 | HDMI-001S | Bottom | 180 | 180 | UNKNOWN | 1 | D2_P | down-right (+4.50, +3.36) | down-left
(-4.50, +3.36) | CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| J3 | C41427493 | USB-C_HX_16P | Top | 180 | 180 | UNKNOWN | A1B12 | GND | down-right (+3.20, +2.55) | - | CH
ECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| J4 | C160405 | JST_SM06B | Bottom | 0 | 0 | UNKNOWN | 1 | PWR_C | up-right (+2.50, -2.00) | up-left (-2.50,
-2.00) | CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| J5 | C41427493 | USB-C_HX_16P | Top | 90 | 90 | UNKNOWN | A1B12 | GND | down-left (-2.55, +3.20) | - | CHECK
IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| J6 | C160407 | JST_SM08B | Top | 90 | 90 | UNKNOWN | 1 | 3V3 | down-left (-2.00, +3.50) | - | CHECK IN PREVI
EW: JLC offset not verified, pin 1 must appear where the two view columns say |
| J7 | C585350 | microSD_Molex_104031-0811 | Bottom | 270 | 90 | 180 | 1 | SD_D2_C | up-left (-5.45, -3.10) |
up-right (+5.45, -3.10) | OK (offset verified in JLC preview 2026-11-07) |
| J8 | C88373 | U.FL-R-SMT-1 | Bottom | 0 | 0 | UNKNOWN | 1 | LNA_OUT | down (+0.00, +1.52) | down (-0.00, +1.
52) | CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| Q1 | C15127 | SOT-23 | Bottom | 0 | 270 | 270 | 1 | SD_PWR_N | down-right (+0.95, +0.94) | down-left (-0.95,
+0.94) | OK (offset verified in JLC preview 2026-11-07) |
| U10 | C92243 | SSOP-4_1.27 | Bottom | 0 | 270 | 270 | 1 | PWR_LED | up-right (+3.17, -0.63) | up-left (-3.1
7, -0.63) | OK (offset verified in JLC preview 2026-11-07) |
| U11 | C92243 | SSOP-4_1.27 | Bottom | 0 | 270 | 270 | 1 | RST_LED | up-right (+3.17, -0.63) | up-left (-3.1
7, -0.63) | OK (offset verified in JLC preview 2026-11-07) |
| U12 | C92243 | SSOP-4_1.27 | Bottom | 0 | 270 | 270 | 1 | LED_IN | up-right (+3.18, -0.64) | up-left (-3.18,
-0.64) | OK (offset verified in JLC preview 2026-11-07) |
| U14 | C138714 | USON-10 | Top | 0 | 0 | UNKNOWN | 1 | USB2_DP | down-left (-1.00, +0.41) | - | CHECK IN PREV
IEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| U15 | C78988 | SOT-23-5 | Top | 90 | 0 | UNKNOWN | 1 | 3V3 | down-right (+1.14, +0.95) | - | CHECK IN PREVIE
W: JLC offset not verified, pin 1 must appear where the two view columns say |
| U16 | C2155674 | WSON-8_2x2_EP | Bottom | 90 | 90 | UNKNOWN | 1 | EF_DVDT | down-right (+0.75, +0.95) | down
-left (-0.75, +0.95) | CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns
```

```
say |
| U2 | C78988 | SOT-23-5 | Bottom | 0 | 270 | 270 | 1 | 5V | down-right (+0.95, +1.14) | down-left (-0.95, +1.
14) | OK (offset verified in JLC preview 2026-11-07) |
| U3 | C138714 | USON-10 | Bottom | 90 | 90 | UNKNOWN | 1 | USB_DN | down-left (-0.41, +1.00) | down-right (+
0.41, +1.00) | CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| U4 | C882825 | SOT-23-5 | Bottom | 0 | 270 | 270 | 1 | 3V3 | down-right (+0.95, +1.14) | down-left (-0.95, +
1.14) | OK (offset verified in JLC preview 2026-11-07) |
| U6 | C201698 | MSOP-8 | Bottom | 90 | 90 | UNKNOWN | 1 | 3V3 | down-left (-2.11, +0.98) | down-right (+2.11,
+0.98) | CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| U7 | C3008606 | BGA-64_6x6 | Bottom | 180 | 180 | UNKNOWN | A1 | TC_REXT | down-left (-2.27, +2.27) | down-r
ight (+2.27, +2.27) | CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns
say |
| U8 | C138714 | USON-10 | Bottom | 180 | 180 | UNKNOWN | 1 | D1_N | up-left (-1.00, -0.41) | up-right (+1.00,
-0.41) | CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| U9 | C138714 | USON-10 | Bottom | 180 | 180 | UNKNOWN | 1 | CK_N | up-left (-1.00, -0.41) | up-right (+1.00,
-0.41) | CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| Y1 | C725995 | OSC3225 | Bottom | 0 | 0 | UNKNOWN | 1 | OSC_VDD | down-right (+1.05, +0.83) | down-left (-1.
05, +0.83) | CHECK IN PREVIEW: JLC offset not verified, pin 1 must appear where the two view columns say |
| C69 | C347552 | CP_7343 |  |  |  |  | +/- | + 5V_DAMP, - GND |  |  | polarity OK |

RESULT PASS: 24 polarized parts, 7 with a verified JLC offset, 17 UNKNOWN (check in JLC preview)
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附錄 20：CPL_pin1_check.md

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# CPL rotation and pin-1 check table (r7d)

r7b M2: one row per placed part: CPL angle, KiCad angle, and where pin 1 must appear in JLC's placement previe
w.

Pin 1 is read from the saved board (pad "1", or A1 for the BGA, or the first pad of the connectors), as an off
set from the part centre in
board coordinates seen from the top (x right, y toward the back wall = down in KiCad and in JLC's top previe
w). For bottom-side parts the
same offset is also given as seen in JLC's bottom preview, which shows the board flipped about the vertical ax
is (x mirrored).
The JLC angle convention is per LCSC part (r6/r7 history: C2871228 at 0 and C78988 at 270 on the same SOT-23-5
land), so nothing here
predicts JLC's angle; the table only says what the preview must show. "verified" means checked by a person in
JLC's preview (2026-11-07).

| Ref | LCSC | Land | Side | KiCad deg | CPL deg | CPL-KiCad | Pin | Pin 1 from centre, top view (mm) | Same p
in in JLC bottom view | Status |
|---|---|---|---|---:|---:|---:|---|---|---|---|
| C69 | C347552 | CP_7343 | Bottom | 0 | 0.0 | 0.0 | 1 | right (+3.10, +0.00) | left (-3.10, +0.00) | CHECK in
preview; new in r7c: polarized (pin 1 = +, silk + beside it) |
| D1 | C48260 | X1SON-2_0.6x1 | Top | 0 | 0.0 | 0.0 | 1 | left (-0.35, +0.00) |  | CHECK in preview |
| J2 | C720616 | HDMI-001S | Bottom | 180 | 180.0 | 0.0 | 1 | down-right (+4.50, +3.36) | down-left (-4.50, +
3.36) | CHECK in preview |
| J3 | C41427493 | USB-C_HX_16P | Top | 180 | 180.0 | 0.0 | A1B12 | down-right (+3.20, +2.55) |  | CHECK in pr
eview |
| J4 | C160405 | JST_SM06B | Bottom | 0 | 0.0 | 0.0 | 1 | up-right (+2.50, -2.00) | up-left (-2.50, -2.00) | C
HECK in preview |
| J5 | C41427493 | USB-C_HX_16P | Top | 90 | 90.0 | 0.0 | A1B12 | down-left (-2.55, +3.20) |  | CHECK in previ
ew |
| J6 | C160407 | JST_SM08B | Top | 90 | 90.0 | 0.0 | 1 | down-left (-2.00, +3.50) |  | CHECK in preview |
| J7 | C585350 | microSD_Molex_104031-0811 | Bottom | 270 | 90.0 | 180.0 | 1 | up-left (-5.45, -3.10) | up-rig
ht (+5.45, -3.10) | CHECK in preview (seen right on 2026-11-07; re-check, review M2) |
| J8 | C88373 | U.FL-R-SMT-1 | Bottom | 0 | 0.0 | 0.0 | 1 | down (+0.00, +1.52) | down (-0.00, +1.52) | CHECK
in preview |
| Q1 | C15127 | SOT-23 | Bottom | 0 | 270.0 | 270.0 | 1 | down-right (+0.95, +0.94) | down-left (-0.95, +0.94)
| CHECK in preview (seen right on 2026-11-07; re-check, review M2) |
| U10 | C92243 | SSOP-4_1.27 | Bottom | 0 | 270.0 | 270.0 | 1 | up-right (+3.17, -0.63) | up-left (-3.17, -0.6
3) | CHECK in preview (seen right on 2026-11-07; re-check, review M2) |
| U11 | C92243 | SSOP-4_1.27 | Bottom | 0 | 270.0 | 270.0 | 1 | up-right (+3.17, -0.63) | up-left (-3.17, -0.6
3) | CHECK in preview (seen right on 2026-11-07; re-check, review M2) |
| U12 | C92243 | SSOP-4_1.27 | Bottom | 0 | 270.0 | 270.0 | 1 | up-right (+3.18, -0.64) | up-left (-3.18, -0.6
4) | CHECK in preview (seen right on 2026-11-07; re-check, review M2); moved 0.08 mm in r7c |
| U14 | C138714 | USON-10 | Top | 0 | 0.0 | 0.0 | 1 | down-left (-1.00, +0.41) |  | CHECK in preview |
| U15 | C78988 | SOT-23-5 | Top | 90 | 0.0 | 270.0 | 1 | down-right (+1.14, +0.95) |  | CHECK in preview |
| U16 | C2155674 | WSON-8_2x2_EP | Bottom | 90 | 90.0 | 0.0 | 1 | down-right (+0.75, +0.95) | down-left (-0.7
5, +0.95) | CHECK in preview |
| U2 | C78988 | SOT-23-5 | Bottom | 0 | 270.0 | 270.0 | 1 | down-right (+0.95, +1.14) | down-left (-0.95, +1.1
4) | CHECK in preview (seen right on 2026-11-07; re-check, review M2) |
| U3 | C138714 | USON-10 | Bottom | 90 | 90.0 | 0.0 | 1 | down-left (-0.41, +1.00) | down-right (+0.41, +1.00)
| CHECK in preview |
| U4 | C882825 | SOT-23-5 | Bottom | 0 | 270.0 | 270.0 | 1 | down-right (+0.95, +1.14) | down-left (-0.95, +1.
14) | CHECK in preview (seen right on 2026-11-07; re-check, review M2) |
| U6 | C201698 | MSOP-8 | Bottom | 90 | 90.0 | 0.0 | 1 | down-left (-2.11, +0.98) | down-right (+2.11, +0.98)
| CHECK in preview |
| U7 | C3008606 | BGA-64_6x6 | Bottom | 180 | 180.0 | 0.0 | A1 | down-left (-2.27, +2.27) | down-right (+2.27,
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+2.27) | CHECK in preview; moved -2.5 mm in x in r7d |
U8	C138714	USON-10	Bottom	180	180.0	0.0	1	up-left (-1.00, -0.41)	up-right (+1.00, -0.41)	CHECK in preview
U9	C138714	USON-10	Bottom	180	180.0	0.0	1	up-left (-1.00, -0.41)	up-right (+1.00, -0.41)	CHECK in preview
Y1	C725995	OSC3225	Bottom	0	0.0	0.0	1	down-right (+1.05, +0.83)	down-left (-1.05, +0.83)	CHECK in preview
C1	C19702	C0603	Top	90	90.0	0.0	1	down (+0.00, +0.78)		non-polar, either way works
C10	C19702	C0603	Bottom	0	0.0	0.0	1	right (+0.78, +0.00)	left (-0.78, +0.00)	non-polar, either way works
C11	C19702	C0603	Bottom	0	0.0	0.0	1	right (+0.78, +0.00)	left (-0.78, +0.00)	non-polar, either way works
C12	C19702	C0603	Bottom	0	0.0	0.0	1	right (+0.78, +0.00)	left (-0.78, +0.00)	non-polar, either way works
C13	C1537	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; new place and rotation in r7d
C14	C19666	C0603	Bottom	90	90.0	0.0	1	down (+0.00, +0.78)	down (-0.00, +0.78)	non-polar, either way works
C15	C1537	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; new place and rotation in r7d
C16	C19666	C0603	Bottom	90	90.0	0.0	1	down (+0.00, +0.78)	down (-0.00, +0.78)	non-polar, either way works; new place and rotation in r7d
C17	C1537	R0402	Bottom	180	180.0	0.0	1	left (-0.48, +0.00)	right (+0.48, +0.00)	non-polar, either way works; new place and rotation in r7d
C18	C19666	C0603	Bottom	0	0.0	0.0	1	right (+0.78, +0.00)	left (-0.78, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
C19	C1537	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
C2	C19702	C0603	Top	90	90.0	0.0	1	down (+0.00, +0.78)		non-polar, either way works
C20	C19666	C0603	Bottom	0	0.0	0.0	1	right (+0.78, +0.00)	left (-0.78, +0.00)	non-polar, either way works; moved 0.05 mm in r7c
C21	C1525	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works; new place and rotation in r7d
C22	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
C23	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
C24	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved 0.05 mm in r7c; moved -2.5 mm in x in r7d
C25	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
C26	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
C27	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
C29	C1525	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
C3	C1525	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
C32	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
C33	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
C4	C1525	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
C40	C19702	C0603	Bottom	0	0.0	0.0	1	right (+0.78, +0.00)	left (-0.78, +0.00)	non-polar, either way works
C41	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
C5	C52923	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
C50	C1525	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
C6	C1525	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works

C60	C45783	C0805	Bottom	90	90.0	0.0	1	down (+0.00, +0.95)	down (-0.00, +0.95)	non-polar, either way works
C61	C45783	C0805	Bottom	90	90.0	0.0	1	down (+0.00, +0.95)	down (-0.00, +0.95)	non-polar, either way works
C62	C45783	C0805	Bottom	90	90.0	0.0	1	down (+0.00, +0.95)	down (-0.00, +0.95)	non-polar, either way works
C63	C1555	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
C64	C52923	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
C65	C45783	C0805	Top	0	0.0	0.0	1	left (-0.95, +0.00)		non-polar, either way works
C66	C45783	C0805	Top	90	90.0	0.0	1	down (+0.00, +0.95)		non-polar, either way works
C67	C45783	C0805	Top	90	90.0	0.0	1	down (+0.00, +0.95)		non-polar, either way works
C68	C1555	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
C70	C385032	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
C71	C1525	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
FB2	C98485	C0805	Bottom	0	0.0	0.0	1	right (+0.95, +0.00)	left (-0.95, +0.00)	non-polar, either way works
FB3	C85840	C0805	Bottom	0	0.0	0.0	1	right (+0.95, +0.00)	left (-0.95, +0.00)	non-polar, either way works
FB4	C98485	C0805	Bottom	0	0.0	0.0	1	right (+0.95, +0.00)	left (-0.95, +0.00)	non-polar, either way works
FB5	C85840	C0805	Bottom	0	0.0	0.0	1	right (+0.95, +0.00)	left (-0.95, +0.00)	non-polar, either way works
FB6	C19330	C0603	Bottom	90	90.0	0.0	1	down (+0.00, +0.78)	down (-0.00, +0.78)	non-polar, either way works
L1	C83423	IND4020	Bottom	0	0.0	0.0	1	right (+1.50, +0.00)	left (-1.50, +0.00)	non-polar, either way works
L2	C83423	IND4020	Top	0	0.0	0.0	1	left (-1.50, +0.00)		non-polar, either way works
R1	C25905	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R10	C25755	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R11	C25744	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
R12	C25744	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
R13	C25744	R0402	Top	90	90.0	0.0	1	down (+0.00, +0.48)		non-polar, either way works
R2	C25905	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R20	C4109	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved -2.5 mm in x in r7d
R21	C25744	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R22	C25879	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R23	C25879	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R24	C25092	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R25	C25744	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R26	C25744	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R27	C25744	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R28	C25744	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works; moved 0.08 mm in r7c
R3	C25905	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R30	C11702	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R32	C25741	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works

R4	C25905	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R40	C25744	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R42	C17168	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R43	C25794	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R44	C25794	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R45	C25794	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R46	C25794	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R47	C25794	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R50	C25755	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R51	C122548	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-pola r, either way works
R52	C25117	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R53	C25117	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R54	C25099	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R55	C25744	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R60	C26083	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R61	C728559	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R62	C319512	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R63	C728559	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R65	C17168	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R66	C25076	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R67	C25794	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works
R68	C247444	C0805	Bottom	0	0.0	0.0	1	right (+0.95, +0.00)	left (-0.95, +0.00)	non-polar, either way works; new in r7c
R71	C25741	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R72	C25792	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R73	C11702	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R74	C25092	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R75	C25092	R0402	Bottom	90	90.0	0.0	1	down (+0.00, +0.48)	down (-0.00, +0.48)	non-polar, either way works
R76	C25092	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R77	C25092	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R78	C25092	R0402	Bottom	0	0.0	0.0	1	right (+0.48, +0.00)	left (-0.48, +0.00)	non-polar, either way works
R79	C25092	R0402	Top	270	270.0	0.0	1	up (+0.00, -0.48)		non-polar, either way works
R80	C320843	R0201	Top	270	270.0	0.0	1	up (+0.00, -0.27)		non-polar, either way works
R81	C320843	R0201	Top	270	270.0	0.0	1	up (+0.00, -0.27)		non-polar, either way works
R9	C25741	R0402	Top	0	0.0	0.0	1	left (-0.48, +0.00)		non-polar, either way works

附錄 21：README.md（打板包說明）

```
# MF KVM（P4 KVM Mini）

40 × 40 mm、6 層單片板的 KVM over IP，目標外殼 44 × 44 × 20 mm。主控是 ESP32-P4-Module 基本款（SKU 28101，內建 ESP32-C6，2.4 GHz，天線訊號從第 2 腳經模組上的 R99 出來，再到板上的 J8 U.FL），影像用 TC358743，韌體是 ESP-KVM。

**狀態：打板包 `fab_r7b/` 只能送 CAM 預審，還不能算放行。** r7b 是 r7 依獨立審查（`REVIEW_independent.md`）修正的版本，改了什麼見 `R7B_REPORT.md`。 ERC、DRC、`verify.py`（含新的電氣規則表）、阻焊幾何、孔到銅、零件外形、阻抗都過了，但這些是檔案層級的檢查，**沒有打過板、沒有焊過、沒有實機驗證**。r7 新加的 1.2 V 降壓（U15）和它的遠端回授線沒有上過電。裸板和雙面組裝都要等 CAM 預審結果與 CPL 旋轉角確認後才放行。

## 打板包（JLCPCB）
送審用 `fab_r7b/p4_kvm_mini_gerber.zip`（雜湊見 `fab_r7b/SHA256.txt`），BOM 和座標用 `fab_r7b/BOM_JLC.csv`、`fab_r7b/CPL_JLC.csv`，下單預覽逐顆對照 `fab_r7b/CPL_pin1_check.md`。r7 的改動見 `R7_REPORT.md`，r7b 的見 `R7B_REPORT.md`。

**不要用 `fab_r7/`。** 1.2 V 分壓是 1% 電阻（AVDD12 最差只剩約 1 mV）、遠端回授線斷開會讓 1.2 V 升到 3.3 V、HDMI +5V 直接接 HPDI、U1 沒有絲印外框、J6 離 U1 焊墊 0.225 mm、TC358743 RESETN 是上拉、不焊零件和客供 U1 的說明不一致。

**不要用 `fab_r6/`。** 它的 U6（PCA9517）EN 接 3V3，低於 EN 的 VIH（0.7 × VCCB = 3.5 V），DDC 中繼不保證會開；Y1 的輸出掛了 2 顆 47 pF，超過振盪器 15 pF 的負載；1.2 V 用 TLV73325 → TLV73312 兩級 LDO，U4 最差負載 98%、U5 在 SOT-23 裡發熱、AVDD12 最差會低於 1.15 V；有 4 組零件在 ±0.1 mm 貼裝誤差下會碰到；KiCad 疊構和下單疊構不一致；CSI_D0 有 4 個 90 度彎，CSI_D0、CSI_CK 線內長度差 2.08、1.47 mm。

更舊的版本（`fab_r5/` 以前）的問題見 `../v2_r6/README.md`，這裡不重複。

| 項目 | 設定 |
|---|---|
| 層數、厚度 | 6 層，1.2 mm 板厚，1 oz 外層 |
| 疊構 | F｜GND｜訊號｜3V3｜GND｜B。**JLC06121H-3313**：外層介質 3313 0.0994 mm、core 0.35 mm、內層 2116 0.1088 mm，銅 0.035 / 0.0152 mm，銅加介質 1.1384 mm。r7 起 KiCad 板檔的疊構就是這組數字，Gerber job 檔跟著一致。介電常數（3313 4.1、2116 4.16、core 4.6）是解算器用的值，不是這個資料夾裡的 JLC 文件 |
| 阻抗 | 場解算（`tools/r6/s3_solver.py`，含阻焊）：TMDS、CSI 0.127/0.198 mm **99.1 Ω**；USB 0.18/0.32 mm **89.1 Ω**；LNA_OUT 0.15 mm、同層鋪銅間隙 0.25 mm **50.5 Ω**。都在 ±10% 內，線寬沒改。另用 Kirschning-Jansen 公式交叉驗證，薄銅條件下差 2%。表在 `fab_r7/impedance_r7.txt`。**沒有跑 JLC 的線上計算器** |
| 最小線寬、線距 | 0.09 mm |
| 最小過孔 | 0.4/0.2 mm，V1V2 有 3 顆 0.5/0.25。選 Epoxy Filled & Capped（JLC 6 層預設）。**不要選阻焊塞孔**：JLC 要求塞孔離開窗 0.35 mm 以上，這塊板不符合 |
| 阻焊 | **綠色**，expansion 0（焊盤與開窗 1:1）。**最小 bridge 0.1088 mm**（J5.1 對 C64.2，同網路），開窗到其他網路銅最小 0.101 mm，明細見 `fab_r7b/mask_report.txt`。（r6 README 寫的「最小 bridge 0.15 mm」是 r5 以前的數字，r6 加了 C64 之後就已經是 0.1088） |
| 孔到銅 | 依 JLC 公開規格：過孔孔 0.20，NPTH 0.20，PTH 外層 0.28、內層 0.30。實測最小：過孔 0.2000（外層）、0.2005（內層），NPTH 0.3005，PTH 0.3017（外層）、0.3025（內層），明細見 `fab_r7/hole_report.txt` |
| 組裝 | 雙面、Standard PCBA。BGA 需要 X-ray |
| 拼板 | 40 mm 太小，依 JLC 要求加工藝邊或拼板 |

**要先處理：**
1. **P4 模組：** BOM 不含 U1，**JLC 不貼，由客戶組裝後手焊**；U1 的 88 個焊盤保留錫膏。買 Waveshare `ESP32-P4-Module` 基本款（SKU 28101）。正面絲印有模組外框、第 1 腳那一角切角和圓點，對準再焊。焊完量 U1.23（I2C_SCL）對 GND、對 3V3 沒有短路（它和 J6.2 GND 相隔 0.425 mm）。**模組上的 R99（0R）要留著**，R75 不要焊。U.FL 天線另外買，接 J8。確認晶片版本，韌體分 rev 3 和舊版。
2. **CPL 旋轉角：** 下單預覽要逐顆確認方向，特別是背面零件。2026-11-07 對過的：U2、U4、Q1、U10、U11、U12 是 270 度，J7 是 90 度（`fab_pack.py` 的 `JLC_ROT`）。**r7 新增或換料、沒有在 JLC 預覽對過的：** U15（SY8089，**正面**；同料號的 U2 在背面是 270 度，正面的 0 度沒對過）、L2、C65–C68、R61–R63、R65、U6（換成 C201698，封裝相同）、R28（移位）。r6 留下的 J8、C64、R60 也還沒對過。U5 拿掉了，它的 0 度（C2871228，JLC 3D 預覽確認過）不再用到。在預覽裡手動旋轉不會存檔，要改 CPL 重傳。
3. **不要焊：** R31、R41、C30、C31（r7b 起這些焊盤沒有錫膏）。**不要同時焊 R41 和 R42**（3V3 對 VO4 短路）。**R65（0
```

R）和 R66（100 Ω）兩顆都要焊**：R65 是遠端回授，R66 是遠端線斷開時的本地備援。見 `fab_r7b/CONSIGNED_AND_NOT_FITTED.csv`。
4. **低庫存：** TC358743 在 JLC 剩 225 顆（`stock.py`，2026-11-08）。
5. **新的擴充料：** U6 TCA9517ADGKR（C201698）；r7b 的 R61、R63 33k 0.1%（C728559）、R62 560 Ω 0.1%（C319512，庫存 7,835）、R51 33.2k（C122548）。R67 51k、R66 100 Ω 是基本庫。U15、L2 和 U2、L1 同料號，不用多一個料站。

供電要求
- **用能供 1.5 A 以上的 Type-C 電源接 J5。** 板子的 CC 只有 5.1 kΩ 下拉，照規範只能當預設電流。
- **電腦的 USB 2.0 埠只保證 500 mA。** r7 電流預算：5V 典型約 384 mA、最差約 1019 mA（r6 是 460 / 1134），見 `R7_REPORT.md` 第 3 項。
- **從電腦燒錄或看 console 時，關掉 Wi-Fi 或降低發射功率。**

1.2 V（r7）
U5（TLV73312）拿掉，1.2 V 改由 U15（SY8089AAAC，和 U2 同料）從 3V3 降壓，放在正面 J2 上方原本只有絲印的空位。L2 2.2 µH（和 L1 同料）、C65 輸入 22 µF、C66/C67 輸出 2 × 22 µF。U4 只剩 AVDD25（最差 19 mA）。
- **遠端回授：** 回授分壓（R61 33k + R62 560 Ω 對 R63 33k，r7b 起都是 0.1%；C68 22 pF）在 U15 旁邊，分壓上端經 V1V2_SNS 一路接到磁珠入口旁的 R65（0R），所以 U15 到磁珠之間約 76 mΩ 的走線壓降被補償。最差情況 AVDD12 在球上 **1.1617–1.2373 V**（規格 1.15–1.25 V），上下各約 12 mV 餘量（r7 用 1% 電阻時只有約 1 mV）。
- **V1V2_SNS 很長（約 51 mm）**，大部分在 In2（上下是 GND 和 3V3 平面），最後約 13 mm 在背面經過 FB5 和 J5 旁邊。回授的穩定度和雜訊沒有量過。
- **斷線保護（r7b）：** R66 100 Ω 把回授節點也接到 U15 本地的 V1V2。兩端都通時只偏 0.09 mV；遠端線或 R65 斷開時，U15 改成在本地取樣，AVDD12 最差掉到約 1.139 V（低於規格但不會過壓）。R61 或 R62 本身開路則沒有保護，1.2 V 會升到 3V3。
- **只用本地取樣（備案）：** 拿掉 R65。AVDD12 最差約 1.139 V。
- **第一片要量：** AVDD12 球上的電壓與雜訊（規格 40 mVpp）、U15 負載瞬變和穩定度、輕載時的漣波。

CAM 預審要確認
1. 接受 1:1 阻焊開窗、不另外加 expansion。若廠內加大，要重新對照 `fab_r7/mask_report.txt`。
2. U7（0.65 mm BGA，0.3 mm 焊盤）在 1:1 開窗下的對位公差可以接受。
3. 過孔用 Epoxy Filled & Capped。
4. 疊構和阻抗：實際介質型號，TMDS、CSI、USB 線寬線距對得到目標阻抗。
5. 0.09/0.09 mm 線寬線距在這個疊構可做，J2、J3、J5 的機構孔和槽孔判讀正確。
6. **逐顆確認 CPL 旋轉角與面別**（雙面組裝），U1 客供流程，以及工藝邊或拼板。
7. 組裝間隙：`fab_r7/body_report.txt` 依規格書外形檢查所有同面零件組合。**r7 起 `bodycheck.py` 把「最差值 < 0」當成錯誤**（最差值 = 最大尺寸再扣每顆 ±0.1 mm 貼裝誤差，**0.1 mm 是假設**，JLC 沒有公開數字）。現在沒有任何組合低於 0，最緊的是 C19/C20 0.000、FB2/C24 0.005、U6/R28 0.006、L1/U12 0.010 mm。TI DBV 外形的毛邊（每邊 0.25 mm）算進去時 U4/Q1 是 +0.055 mm。**Q1 的外形不是原廠圖**（AOS 沒有外形頁，暫用 TI DBZ0003A）。
8. 孔位餘量：過孔孔到走線最小 0.2000 mm，剛好等於限值。`holecheck.py` 沒有檢查孔到孔。
9. J5 A1B12 的 GND 散熱連接只有 1 條（DRC 警告 starved_thermal），可焊性要確認。
10. 絲印：MF KVM 和 R7B（1.0 mm）直排在 J2 和 J3 之間（R67 占了下半段）；U1 外框、切角、第 1 腳圓點（r7b）；JLC 公開規格是字高 1.0 mm、線寬 0.15 mm 以上，P4 MINI R7 剛好在下限。測試點標籤（0.5 mm）低於規格，可能印不清楚。

Courtyard 警告
DRC 有 18 個 courtyard 重疊警告，全部是外框加 0.1 mm 邊界造成的，零件本體沒有重疊（`body_report.txt`）。最緊的幾組（最大尺寸 / 最差，mm）：C19/C20 0.200 / 0.000、U6/R28 0.206 / 0.006、L1/U2 0.280 / 0.080、C10/Q1 0.375 / 0.094、Q1/R43–R45 0.300 / 0.100、C20/U8 0.300 / 0.100、C64/U2 0.350 / 0.150。TP6/J4 是測試點外框邊界，TP 沒有本體。C30/U3 的 C30 不焊。

新光耦（需要實機測試）
沒有改動，見 `../v2_r6/README.md` 同名一節：ATX 開機和重置要實機測試過才算數。

接線
接頭	用途
J2 HDMI、J3 USB-C	接主機，在 I/O 面。J3 是 P4 的 USB OTG-HS，主機看到的鍵盤、滑鼠和虛擬媒體都走這裡；不供電給板子，也不是燒錄或 console
J5 USB-C	5V 供電，也是燒錄口（USB-Serial-JTAG，GPIO24/25）和**次要** console（只有應用程式 log）
J7 microSD	站在接線面外看從右牆插入，推拉式
J6 SM08B	圓形螢幕：1 3V3、2 GND、3 DIN、4 CLK、5 CS、6 DC、7 RST、8 BL（BL 接 3V3，背光常亮）。r7b 往 +x 移 0.2 mm，焊盤離 U1 焊盤 0.425 mm。見下
J4 SM06B	前面板：PWR C/E、RST C/E、LED K/A，在背面

| J8 U.FL | 天線，在背面靠接線面的對角，接 FPC 天線 |
| TP1–TP8 | 底面：BOOT、EN、UART0 TX/RX、C6 RX/TX/BOOT、GND。**主要 console 是這裡的 UART0（GPIO37/38，115200）** |

J6 和 Waveshare 1.28inch LCD Module（SKU 19192）： wiki 的接腳順序是 VCC、GND、DIN、CLK、CS、DC、RST、BL，工作電壓 3.3 V / 5 V，模組上是 PH2.0 8P（隨附 PH2.0 8P 20 cm 線）。J6 的 1–8 腳順序和它一樣，所以 **SH1.0 8P 對 PH2.0 8P 一對一、不交叉** 的線可以用：J6 第 1 腳（3V3）接模組 VCC，第 8 腳（BL）接 BL。**wiki 沒寫模組端 PH2.0 的腳號和哪一邊是 VCC**，做線前要看模組絲印確認 VCC 那一腳。背光電流 wiki 沒有寫，電流預算沿用 r6 的假設（30 / 60 mA）。這次沒改 J6。

腳位：TC358743 I2C 7/8、RESET 23（**r7b 起 R21 是下拉**：韌體要把 GPIO23 拉高才會放開 reset，現行韌體的 `bridge_resetn_pulse()` 會做，見 `firmware/BUILD_v2.md` 最後一節）；HDMI +5V 經 R67 51k 接 HPDI；LCD DIN 9、CLK 10、CS 11、DC 12、RST 13；ATX PWR 4、RST 5、LED 2；TF 39–45；GPIO3 是主機 VBUS 分壓，韌體尚未使用。

韌體

r7 沒有改韌體，韌體相關事項（`firmware/p4_kvm_mini.defaults`、ATX 預設關閉、VO4 必須設 3.3 V 才關得掉 Q1、晶片版本等）見 `../v2_r6/README.md` 的「韌體」一節，內容仍然適用。r7 的硬體改動不影響腳位。

重新產生

r7b 的布線檔是 `out/final.ses`（同 `out/final_r7b.ses`）。一般重建只要這幾行：

```
sh tools/check.sh out/final.ses             # 由布線產生板子和電路圖，跑 ERC、DRC
python tools/stock.py out                   # JLC 型號與庫存
python tools/verify.py out data/antmicro/a.net > out/verify.out   # 含電氣規則表、maskcheck、holecheck、bodycheck
python tools/fab_pack.py out fab_r7b        # 任何檢查沒過就拒絕出檔；目標已有打板包也拒絕
python tools/r7/cpl_table.py out fab_r7b/CPL_JLC.csv fab_r7b/CPL_pin1_check.md   # 每顆零件的 CPL 角度與預覽中第 1 腳位置
python tools/r7/impedance_r7.py out/p4_kvm_mini.kicad_pcb        # 從板檔讀疊構，算阻抗
python tools/r7/budget_r7.py out/p4_kvm_mini.kicad_pcb           # 1.2 V 和電流預算
python tools/r6/pairs.py out/p4_kvm_mini.kicad_pcb               # 差動對長度、skew、90 度彎
```

r7 布線是怎麼做出來的（不必重做）：`gen_mini.py` 的電源平面逃線搜尋仍在 r6 的零件位置上跑，r7 的改動在搜尋之後手動加。r6 板子上全部已布的銅以 `R7_KEEP=<板檔>` 交給 Freerouting 當成保護線，Freerouting 只補 r7 新的連線（`--router.fanout.enabled=false`）。步驟和每一輪加了什麼銅見 `R7_REPORT.md`「布線」一節，`tools/r7/ses_new.py` 可以列出任何 SES 裡新加的銅。

外殼

沿用 r6 的外殼檔（`case/stl/`），**尚未試印**。對 r7b 板子重跑 `case/check_case.py` 的結果見 `fab_r7b/case_check.txt`（J6 往 +x 移後，M2 螺絲到 J6 的間隙 0.68 mm）。新加的 U15、L2（高 2.0 mm）在正面 J2 上方，r6 外殼的這一區沒有其他零件。

參考來源

- Antmicro hdmi-mipi-bridge（Apache-2.0）：TC358743 周邊電路。`verify.py` 用的網表和授權聲明在 `data/antmicro/`。
- 規格書：TI TCA9517A（SCPS245E，`datasheets_r7/tca9517a.pdf`）、YXC YSO110TR（`datasheets_r7/dl/C725995_ds.pdf`）、Silergy AN_SY8089/A Rev 0.9A、Toshiba TC358743XBG、TI TLV733P（`../pdf_20261108/規格書/`）。
- Waveshare：ESP32-P4-Module 焊盤圖、STEP、腳位表；1.28inch LCD Module wiki（https://www.waveshare.com/wiki/1.28inch_LCD_Module）。
- KiCad 官方封裝庫：microSD、JST SH、SO-4、SOT、VSSOP、振盪器、電感。

附錄 22：SHA256.txt

```
5edbd11614de56c81819c0f49b667f6346e41d8fef006e6f3f2256d9243e99f7  p4_kvm_mini_gerber.zip
```

附錄 23：SHA256_extra.txt

```
d42e497c69ae9cb69706e73b07c65c926349ad358943ead8b5981732db77ad3e *p4_kvm_mini_layout.pdf
382c7b376db88e8bbea7218c67c0eb604d37f1c365fe97bbe1c962985edf435c *coldstart_r7f.txt
c6b9cc3b94d904f84ca6b6e9db24861db3b5a2608f23a81c3a57cc0cd55f2726 *case_spec_r7f.json
4b45bc54ea82b271f0e0f2c0b15e4ba6ccf31c724af7cde3371484ad550b087b *hand_solder_gap_r7f.txt
1196850c16a4b2a157a04a7c45decab24d31325037ecc55fb3e85ff81ac302cd *csi_via_r7f.txt
bb53fb85be623d0638002a29f294ff73ef3ed9586612187f9551c36a121e96bb *CPL_polarity_pin1_r7f.md
e50eb6b27ba369fa0ef7e270bd4e5dbea1ad79c75ae34cd34259d10a27626784 *refplane_r7f.txt
661233f621cbc8722ea980274b69e6736223043735be01048a719e32365304ac *pairgap_r7f.md
74c02f19aaf42d0fd97e8c94db33fa648de8afa7222147f0c80f565d4df9cf5f *drc_instances_r7f.txt
ba2b49d8ab2cb5f921c4202314adb1f37aabdcbf3c43b33deb416343e0893452 *bodycheck_r7f.txt
5f5a6b257ff74361ad48001facaca34789ca328497c93926829a782363852143 *s3_solver_test_r7f.txt
```